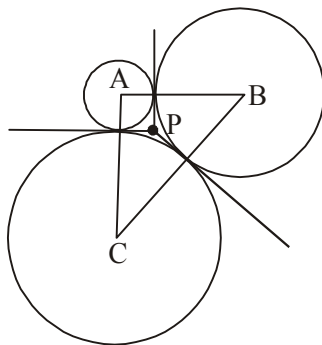


Paragraph for Question T03/3/204 to T03/3/206

Three circles with radii 1,2,3 units and centres A,B,C respectively are drawn to touch each other externally. Now $\triangle ABC$ is formed by joining the centres of the circles and tangents are drawn to each pair of circles at their common point of contact to meet at the point P (as shown in the figure)

Let I_1, I_2, I_3 are the centres of excircles opposite to vertices A,B,C respectively.

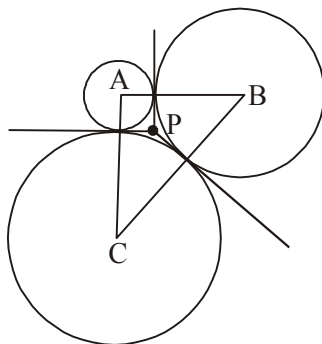
On the basis of above information, answer the following questions :



तीन वृत्त जिनकी त्रिज्यायें 1,2,3 इकाई तथा केन्द्र क्रमशः A,B,C है एक दूसरे को बाह्य स्पर्श करते हैं। वृत्तों के केन्द्रों को जोड़कर त्रिभुज $\triangle ABC$ बनाया जाता है तथा वृत्तों के प्रत्येक युग्म पर उनके उभयनिष्ठ स्पर्श बिन्दु पर स्पर्श रेखायें खींची जाती हैं जो बिन्दु P पर मिलती हैं (चित्रानुसार)

माना I_1, I_2, I_3 क्रमशः शीर्षों A,B,C के सम्मुख बाह्यवृत्तों के केन्द्र हैं।

उपरोक्त जानकारी के आधार पर निम्न प्रश्नों के उत्तर दीजिए।



T03/3/204. Distance between incentre & orthocentre for $\triangle ABC$ is equal to :

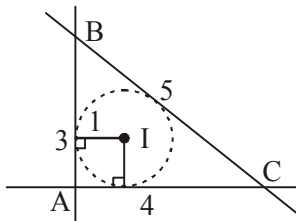
- (A) 1 unit (B) $\sqrt{2}$ units (C) 2 units (D) 3 units

त्रिभुज $\triangle ABC$ के अन्तः केन्द्र तथा लम्बकेन्द्र के मध्य दूरी होगी :

- (A) 1 इकाई (B) $\sqrt{2}$ इकाई
(C) 2 इकाई (D) 3 इकाई

Ans. B

Sol. In $\triangle ABC \Rightarrow AB = 3, BC = 5$ and $AC = 4$
 $\Rightarrow$ It is a right angled triangle $\Rightarrow r = 1$



Distance between incentre & orthocentre is equal to $\sqrt{2}$

T03/3/205. Circumradius of $\triangle I_1 I_2 I_3$ is equal to :

- (A) 3 units (B) 4 units (C) 5 units (D) 6 units

त्रिभुज $\triangle I_1 I_2 I_3$ की परित्रिज्या होगी -

- (A) 3 इकाई (B) 4 इकाई (C) 5 इकाई (D) 6 इकाई

Ans. C

Sol. Circumradius of $\triangle I_1 I_2 I_3 = 2R = 2 \times \frac{5}{2} = 5$

T03/3/206. Perpendicular distance of orthocentre of $\triangle I_1 I_2 I_3$, from side BC is equal to :

- (A) 1 units (B) 2 units (C) 3 units (D) 4 units

त्रिभुज $\triangle I_1 I_2 I_3$ के लम्बकेन्द्र की भुजा BC से लम्बवत् दूरी होगी :

- (A) 1 इकाई (B) 2 इकाई (C) 3 इकाई (D) 4 इकाई

Ans. A

Sol. Orthocentre of $\triangle I_1 I_2 I_3$ will coincide with incentre of $\triangle ABC$, so its distance from BC = 1

INTEGER TYPE QUESTIONS

D05/4/207. If $f(x) = \left(\tan \left(\frac{\pi}{4} + \ln x \right) \right)^{\frac{1}{2} \log_x e}$ is to be made continuous at $x = 1$, then $[f(1)]$ should be equal to

(where $[.]$ denotes greatest integer function) _____.

यदि $f(x) = \left(\tan\left(\frac{\pi}{4} + \ln x\right) \right)^{\frac{1}{2} \log_x e}$ को $x = 1$ पर सतत् बनाया जाता है, तब $[f(1)]$ का मान होगा

(जहाँ $[.]$ महत्तम पूर्णांक फलन को दर्शाता है) _____.

Ans. 2

Sol. $f(x) = \left(\tan\left(\frac{\pi}{4} + \ln x\right) \right)^{\frac{1}{2 \ln x}}$ (1^∞ form)

$$\lim_{x \rightarrow 1} f(x) = e^{\lim_{x \rightarrow 1} \frac{1}{2 \ln x} \left(\tan\left(\frac{\pi}{4} + \ln x\right) - 1 \right)} = e^{\lim_{x \rightarrow 1} \frac{1}{2 \ln x} \left(\frac{2 \tan(\ln x)}{1 - \tan(\ln x)} \right)} = e$$

D05/4/208. The least integral value of a for which the function $f(x) = \left[\frac{(x-2)^3}{a} \right] \sin(x-2) + a \cos(x-2)$, where $[.]$ denotes the greatest integer function, is continuous in $[0, 2]$ is _____.

a का न्यूनतम पूर्णांक मान जिसके लिए फलन $f(x) = \left[\frac{(x-2)^3}{a} \right] \sin(x-2) + a \cos(x-2)$ अन्तराल $[0, 2]$ में सतत्

हैं, (जहाँ $[.]$ महत्तम पूर्णांक फलन को प्रदर्शित करता है) होगा _____.

Ans. 9

Sol. $\sin(x-2)$ and $\cos(x-2)$ are continuous for all x . Since $[x^3]$ is not continuous at integral values of x^3 , $f(x)$ is

continuous in $[0, 2]$ if $\left[\frac{(x-2)^3}{a} \right] = 0, \forall x \in [0, 2]$.

Now, $(x-2)^3 \in [0, 8]$ for $x \in [4, 6]$

$$\Rightarrow a > 8 \text{ for } \left[\frac{(x-2)^3}{a} \right] = 0$$

D05/4/209. Let $f(x) = x^2 + ax + 3$ and $g(x) = x + b$, where $F(x) = \lim_{n \rightarrow \infty} \frac{f(x) + (x^2)^n g(x)}{1 + (x^2)^n}$. If $F(x)$ is continuous at x

= 1 and $x = -1$ then find the value of $(a^2 + b^2)$ _____.

माना $f(x) = x^2 + ax + 3$ तथा $g(x) = x + b$ हैं, जहाँ $F(x) = \lim_{n \rightarrow \infty} \frac{f(x) + (x^2)^n g(x)}{1 + (x^2)^n}$ । यदि $F(x)$, $x = 1$ तथा $x =$

-1 पर सतत् है तब $(a^2 + b^2)$ का मान ज्ञात कीजिए _____.

Ans. 17

Sol. $F(x) = g(x)$ $x > 1$

$$= \frac{f(x) + g(x)}{2} \quad x = 1$$

$$= f(x) \quad -1 < x < 1$$

$$= \frac{f(x) + g(x)}{2} \quad x = -1$$

$$= g(x) \quad x < -1$$

if $F(x)$ is continuous at $x = 1$

$$F(1^+) = F(1) = F(1^-)$$

$$b = a + 3$$

if $F(x)$ is continuous at $x = -1$

$$F(-1^-) = F(-1) = F(-1^+)$$

$$a + b = 5$$

D05/4/210. If $f(x) = x \left(\frac{\sqrt{\{x\}} + \sqrt{[x]+1}}{1 + \sqrt{\{x\}}} \right)$ (wherever defined), then $f'(0^+)$ is :

(where $[.]$ denotes greatest integer function & $\{.\}$ fractional part of function) _____.

यदि $f(x) = x \left(\frac{\sqrt{\{x\}} + \sqrt{[x]+1}}{1 + \sqrt{\{x\}}} \right)$ (जहाँ भी परिभाषित) हो, तो $f'(0^+)$ का मान होगा :

(जहाँ $[.]$ महत्तम पूर्णांक फलन तथा $\{.\}$ भिन्नात्मक भाग फलन को दर्शाता है) _____.

Ans. 1

Sol. $f(0^+) = \lim_{n \rightarrow 0} n \left(\frac{\sqrt{n} + 1}{1 + \sqrt{n}} \right) = 0$

$$f(0^-) = \lim_{n \rightarrow 0} -n \frac{(\sqrt{n+1} + 0)}{(1 + \sqrt{n+1})} = 0 = f(0)$$

$$f'(0^-) = \lim_{n \rightarrow 0} \frac{-n \frac{(\sqrt{n+1} + 0)}{(1 + \sqrt{n+1})}}{-n} = \frac{1}{2}$$

$$f'(0^+) = \lim_{n \rightarrow 0} \frac{n \frac{(\sqrt{n+1})}{1 + \sqrt{n+1}}}{n} = 1$$

D05/4/211. If a and b are the numbers of points of non differentiability of $f(x) = [\sin^{-1}x]$ and $g(x) = \left[\frac{2}{1+x^2} \right], x \geq 0$, (where $[.]$ represents greatest integer function) respectively, then the value of $a + b$ is _____.

यदि a तथा b, $f(x) = [\sin^{-1}x]$ तथा $g(x) = \left[\frac{2}{1+x^2} \right], x \geq 0$, के बिन्दुओं की संख्या (जहां $[.]$ महत्तम पूर्णांक फलन को प्रदर्शित करता है) है जहां ये फलन अवकलनीय नहीं है तो $a + b$ का मान होगा _____.

Ans. 5

Sol. $\sin^{-1}x$ is a monotonically increasing function. Hence, $f(x) = [\sin^{-1}x]$ is discontinuous, where $\sin^{-1}x$ is an integer.

$$\Rightarrow \sin^{-1}x = -1, 0, 1 \quad \text{or} \quad x = -\sin 1, 0, \sin 1$$

$\frac{2}{1+x^2}, x \geq 0$, is a monotonically decreasing function.

Hence, $f(x) = \left[\frac{2}{1+x^2} \right], x \geq 0$ is discontinuous when $\frac{2}{1+x^2}$ is an integer.

$$\Rightarrow \frac{2}{1+x^2} = 1, 2 \Rightarrow x = -1, 0$$

D05/4/212. The number of non differentiability points of function $f(x) = \min \left([x], \{x\}, \left| x - \frac{3}{2} \right| \right)$ for $x \in (0, 2)$, where $[.]$ and $\{.\}$ denote greatest integer function and fractional part function respectively _____.

फलन $f(x) = \min\left([x], \{x\}, \left|x - \frac{3}{2}\right|\right)$ के $x \in (0, 2)$ में अन-अवकलनीयता बिन्दुओं की संख्या होगी, जहाँ $[.]$ तथा $\{.\}$

क्रमशः महत्तम पूर्णांक फलन तथा भिन्नात्मक भाग फलन को दर्शाते हैं _____.

Ans. 3

Sol. $f(x) = [x] \quad 0 < x < 1$

$$= \{x\} \quad 1 \leq x < \frac{5}{4}$$

$$= \left|x - \frac{3}{2}\right| \quad \frac{5}{4} \leq x < 2$$

No. of points where $f(x)$ is non differentiable are three.

$$x = 1, \frac{5}{4}, \frac{3}{2}$$

D05/4/213. Let $g(y) = \lim_{x \rightarrow y} \frac{\tan x - \tan y}{1 - \frac{x}{y} + \left(1 - \frac{x}{y}\right) \tan x \cdot \tan y}$ and $f(x) = x^2$. If $h(x) = \min(f(x), g(x))$, find the number of

points where $h(x)$ is non-derivable _____.

माना $g(y) = \lim_{x \rightarrow y} \frac{\tan x - \tan y}{1 - \frac{x}{y} + \left(1 - \frac{x}{y}\right) \tan x \cdot \tan y}$ तथा $f(x) = x^2$. If $h(x) = \min(f(x), g(x))$ हैं, उन बिन्दुओं की संख्या

ज्ञात कीजिए जहाँ $h(x)$ अन-अवकलनीय है _____.

Ans. 2

Sol. $g(y) = \lim_{x \rightarrow y} \frac{\tan x - \tan y}{1 - \frac{x}{y} + \left(1 - \frac{x}{y}\right) \tan x \cdot \tan y} = \lim_{x \rightarrow y} \left(\frac{\tan x - \tan y}{1 + \tan x \cdot \tan y} \right) \frac{1}{\left(1 - \frac{x}{y}\right)} = \lim_{x \rightarrow y} \frac{y \tan(x - y)}{y - x} = -y$

$$h(x) = \min(f(x), g(x)) = \min(x^2, -x)$$

$$= -x \quad x \leq -1$$

$$= x^2 \quad -1 < x < 0$$

$$= -x \quad x \geq 0$$

$h(x)$ is non-derivable at $x = -1, 0$

D05/4/214. If $f(x) = \begin{cases} \cos x^3; & x < 0 \\ \sin x^3 - |x^3 - 1|; & x \geq 0 \end{cases}$

then find the number of points where $g(x) = f(|x|)$ is non-differentiable _____.

यदि $f(x) = \begin{cases} \cos x^3; & x < 0 \\ \sin x^3 - |x^3 - 1|; & x \geq 0 \end{cases}$

तब उन बिन्दुओं की संख्या होगी जहाँ $g(x) = f(|x|)$ अन-अवकलनीय है _____.

Ans. 2

Sol.
$$\begin{aligned} g(x) &= \sin x^3 - x^3 + 1 & x \geq 1 \\ &= \sin x^3 + x^3 - 1 & 0 \leq x < 1 \\ &= -\sin x^3 - x^3 - 1 & -1 \leq x < 0 \\ &= -\sin x^3 + x^3 + 1 & x \leq -1 \end{aligned}$$

Function is not differentiable at $x = -1, 1$

D05/4/215. Let $f: \mathbb{R}^+ \longrightarrow \mathbb{R}$ be a differentiable function satisfying

$$f(x) = \frac{f(x)}{y} + \frac{f(y)}{x} \quad \forall x, y \in \mathbb{R}^+ \quad \text{also} \quad f(1) = 0; f'(1) = 1$$

find $\lim_{x \rightarrow e} \left[\frac{1}{f(x)} \right]$ (where $[.]$ denotes greatest integer function). _____.

माना $f: \mathbb{R}^+ \longrightarrow \mathbb{R}$ एक अवकलनीय फलन है जो $f(x) = \frac{f(x)}{y} + \frac{f(y)}{x} \quad \forall x, y \in \mathbb{R}^+$ को संतुष्ट करता है।

$$f(1) = 0; f'(1) = 1$$

$\lim_{x \rightarrow e} \left[\frac{1}{f(x)} \right]$ का मान ज्ञात कीजिए (जहाँ $[.]$ महत्तम पूर्णांक फलन को दर्शाता है) _____.

Ans. 2

Sol.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{f\left(x\left(1+\frac{h}{x}\right)\right) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\frac{f(x)}{1+\frac{h}{x}} + \frac{f\left(1+\frac{h}{x}\right)}{x} - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(x)\left(-\frac{h}{x}\right) + f\left(1+\frac{h}{x}\right)}{h\left(1+\frac{h}{x}\right)} = \frac{-f(x)}{x} + \lim_{h \rightarrow 0} \frac{f\left(1+\frac{h}{x}\right) - f(1)}{x^2\left(\frac{h}{x}\right)} \quad (\text{as } f(1) = 0)$$

$$f'(x) = \frac{-f(x)}{x} + \frac{f'(1)}{x^2}$$

$$xf'(x) + f(x) = \frac{1}{x}$$

$$\frac{d}{dx}(xf(x)) = \frac{1}{x}$$

$$xf(x) = \int \frac{1}{x} dx$$

$$xf(x) = \ln x + k$$

put $x=1$, we get $k=0$

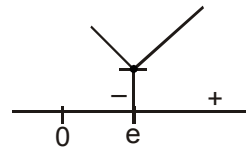
$$\therefore f(x) = \frac{\ln x}{x}$$

$$H(x) = \frac{1}{f(x)} = \frac{x}{\ln x}$$

$$H'(x) = \frac{\ln x \cdot 1 - 1}{(\ln x)^2} \quad H(x) \geq e$$

$$H(e) = e$$

$$\therefore \lim_{x \rightarrow e} \left[\frac{1}{f(x)} \right] = 2$$



D05/4/216. Let $f(x) = \begin{cases} \frac{x}{2} - 1 & 0 \leq x < 1 \\ \frac{1}{2} & 1 \leq x \leq 2 \end{cases}$ and $g(x) = (2x+1)(x-k)+3, x \geq 0$. Then $g(f(x))$ is continuous at $x=1$ then

माना $f(x) = \begin{cases} \frac{x}{2} - 1 & 0 \leq x < 1 \\ \frac{1}{2} & 1 \leq x \leq 2 \end{cases}$ तथा $g(x) = (2x + 1)(x - k) + 3, x \geq 0$ हैं। तब $g(f(x))$, $x = 1$ पर सतत् है तो $12k$

का मान बराबर होगा _____.

Ans. 6

Sol. $g(f(x)) = \begin{cases} g\left(\frac{x}{2} - 1\right) & 0 \leq x < 1 \\ g\left(\frac{1}{2}\right) & 1 \leq x \leq 2 \end{cases}$

$$= \begin{cases} \frac{(x-1)(x-2-2k)}{2} + 3 & 0 \leq x < 1 \\ 4 - 2k & 1 \leq x < 2 \end{cases}$$

$$\lim_{x \rightarrow 1^-} g(f(x)) = 3, g(f(1)) = 4 - 2k$$

$$\text{And } \lim_{x \rightarrow 1^+} g(f(x)) = 4 - 2k$$

For $g(f(x))$ to be continuous at $x = 1$, $4 - 2k = 3$ or $k = \frac{1}{2}$

D05/4/217. Number of points of discontinuity of $f(x) = [x^3 + 3x^2 + 7x + 2]$, where $[.]$ represents the greatest integer function in $[0, 1]$ is _____.

$f(x) = [x^3 + 3x^2 + 7x + 2]$ के असततता बिन्दुओं की संख्या होगी, जहाँ $[.]$, $[0, 1]$ में महत्तम पूर्णांक फलन को दर्शाता है _____.

Ans. 11

Sol. $g(x) = x^3 + 3x^2 + 7x + 2$

$$g'(x) = 3x^2 + 6x + 7 > 0 \text{ for all real } x \text{ as its discriminant is negative.}$$

Thus, $g(x)$ is an increasing function.

$$g(0) = 2 \text{ and } g(1) = 1 + 3 + 7 + 2 = 13$$

Thus, $f(x) = [g(x)]$ is discontinuous when $g(x) = 3, 4, \dots, 13$.

So there are 11 points of continuity.

D05/4/218. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function, where $\mathbb{R}$ is the set of real numbers. For each statement below, write whether it is TRUE or FALSE.

- (A) If $|f(x) - f(y)| \leq 39|x - y|$ for all x, y then f must be continuous everywhere.
- (B) If $|f(x) - f(y)| \leq 39|x - y|$ for all x, y then f must be differentiable everywhere.
- (C) If $|f(x) - f(y)| \leq 39|x - y|^2$ for all x, y then f must be differentiable everywhere.
- (D) If $|f(x) - f(y)| \leq 39|x - y|^2$ for all x, y then f must be constant.

माना $f: \mathbb{R} \rightarrow \mathbb{R}$ एक फलन है, जहाँ $\mathbb{R}$ वास्तविक संख्याओं का समुच्चय है। नीचे दिए गए प्रत्येक कथन के लिए बताइये कि कथन सत्य या असत्य है।

- (A) यदि $|f(x) - f(y)| \leq 39|x - y| \forall x, y$ तो f सभी जगह सतत् होना चाहिए।
- (B) यदि $|f(x) - f(y)| \leq 39|x - y| \forall x, y$ तो f सभी जगह अवकलनीय होना चाहिए।
- (C) यदि $|f(x) - f(y)| \leq 39|x - y|^2 \forall x, y$ तब f सभी जगह अवकलनीय होना चाहिए।
- (D) यदि $|f(x) - f(y)| \leq 39|x - y|^2 \forall x, y$ तो f नियत होना चाहिए।

Ans. TFFT

Sol. In parts a and b, we have $|f(x) - f(\alpha)|$ sandwiched between $\pm 39|x - \alpha|$. As $x \rightarrow \alpha$, $\pm 39|x - \alpha| \rightarrow 0$ and hence $f(x) - f(\alpha) \rightarrow 0$, so f is continuous. But it need not be differentiable, e.g. $f(x) = |x|$ satisfies $f(x) - f(y) = |x| - |y| \leq |x - y| \leq 39|x - y|$. But f is not differentiable at 0.

In parts c and d, we have $\left| \frac{f(x) - f(\alpha)}{x - \alpha} \right| \leq 39|x - \alpha|$, so by reasoning as for part a, we have

$$\lim_{x \rightarrow \alpha} \frac{f(x) - f(\alpha)}{x - \alpha} = 0, \text{ i.e., } f'(\alpha) = 0 \text{ for all } \alpha, \text{ so } f \text{ is a constant function.}$$

D05/4/219. Let A, B and C be unknown constants. Consider the function $f(x)$ defined by

$$f(x) = Ax^2 + Bx + C \text{ when } x = 0,$$

$$= \ln(5x + 1) \text{ when } x > 0.$$

Write the values of the constants A, B and C such that $f''(x)$, i.e., the double derivative of f , exists for all real x . If this is not possible, write “not possible”. If some of the constants cannot be uniquely determined, write “not unique” for each such constant.

माना A, B तथा C अज्ञात नियतांक है। फलन $f(x)$ पर विचार करें जो इस प्रकार है

$$f(x) = Ax^2 + Bx + C \text{ जब } x = 0,$$

$$= \ln(5x + 1) \text{ जब } x > 0.$$

Ans. $A = -\frac{25}{2}, B = 5, C = 0$

Sol. The only problem is at $x = 0$. For continuity, $\ln(0 + 1) = C$. For $f'(0)$ to exist, f must be continuous and the left and right derivatives of f at $x = 0$ (which are easily seen to exist) must match, i.e. $5 = B$. For $f''(0)$ to exist, $f'(0)$ must exist and left and right derivatives of f' at $x = 0$ must match, i.e. $2A = -5^2$. So $A = -\frac{25}{2}, B = 5, C = 0$.

D05/4/220. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a twice differentiable function, where $\mathbb{R}$ denotes the set of real numbers. Suppose that for all real numbers x and y , the function f satisfies

$$f'(x) - f'(y) \leq 3|x - y|$$

Answer the following questions. No credit will be given without full justification.

(a) Show that for all x and y , we must have $|f(x) - f(y) - f'(y)(x - y)| \leq 1.5(x - y)^2$.

(b) Find the largest and smallest possible values for $f''(x)$ under the given conditions.

माना $f: \mathbb{R} \rightarrow \mathbb{R}$ एक द्विअवकलनीय फलन है जहाँ $\mathbb{R}$, वास्तविक संख्याओं के समुच्चय को दर्शाता है। मान लीजिए सभी वास्तविक संख्याओं x तथा y के लिए, फलन f , $f'(x) - f'(y) \leq 3|x - y|$ को संतुष्ट करता है।

निम्नलिखित प्रश्नों के उत्तर दीजिए।

(a) दर्शाइये कि सभी x तथा y के लिए $|f(x) - f(y) - f'(y)(x - y)| \leq 1.5(x - y)^2$ होगा।

(b) $f''(x)$ के महत्तम और न्यूनतम संभव मान ज्ञात कीजिए।

Ans. (b) 3, -3

Sol. (a) Note that the given inequality stays valid if we take absolute value of the LHS, because we may interchange x and y without affecting RHS.

Fix x, y and let $t = x - y$. For now let $x \geq y$, i.e. $t \geq 0$. For $h \in [0, t]$, the value of $y + h$ varies between y and x . We are given that $|f'(y + h) - f'(y)| \leq 3|h|$. Integrate with respect to h over the interval $[0, t]$ to get

$\int_0^t |f'(y + h) - f'(y)| dh \leq \int_0^t 3|h| dh = 1.5t^2$. The LHS should remind us of the following general fact: the absolute value of a definite integral $\leq$ the definite integral of absolute value the same function over the same interval.

So we get $\left| \int_0^t f'(y + h) - f'(y) dh \right| \leq \int_0^t |f'(y + h) - f'(y)| dh$. Combining with the previous

inequality we have $\left| \int_0^t f'(y + h) - f'(y) dh \right| \leq 1.5t^2$. Finally we calculate the LHS.

$\left| \int_0^t f'(y + h) dh - \int_0^t f'(y) dh \right| = |f(y + t) - f(y) - f'(y)t|$, where the first integral is calculated using the fundamental theorem of calculus and the second one is just the integral of the constant $f'(y)$. Substituting $x - y$ for t gives the desired result.

Notes: (1) If $x < y$, then $t < 0$. We use the same strategy but all definite integrals should be taken over $[t, 0]$.

Now $\int_t^0 3|h|dh = 1.5t^2 = \text{LHS of the desired inequality}$. In the final calculation we get

$\int_t^0 f'(y+h) - f'(y) dh = f(y) - f(y+t) + f'(y)t = \text{negative of the previous answer}$. So when we take absolute value of this integral, we again get the same RHS of the desired inequality. (2) $f'(y+h) - f'(y)$ and its absolute value are integrable functions of h because they are continuous. This is because $f'(y+h)$ is just a shift of the function f' and f' is continuous because it is differentiable by hypothesis.

(b) We have, for $x \neq y$, $\left| \frac{f'(x) - f'(y)}{x - y} \right| \leq 3$, so $-3 \leq \frac{f'(x) - f'(y)}{x - y} \leq 3$. Taking limit as $y \rightarrow x$, we get $-3 \leq f''(x) \leq 3$. It is easy to provide examples where f'' attains the extreme values ± 3 , e.g., $f(x) = \pm 1.5x^2$. These satisfy the hypothesis and have constant $f'' = \pm 3$

D05/4/221. A function $f(x)$ is defined by the following formulas

$$f(x) = \begin{cases} x^2 + 1 & \text{when } x \text{ is irrational,} \\ \tan(x) & \text{when } x \text{ is rational.} \end{cases}$$

At how many x in the interval $[0, 4\pi]$ is $f(x)$ continuous?

एक फलन $f(x)$ इस प्रकार परिभाषित है कि

$$f(x) = \begin{cases} x^2 + 1 & \text{जब } x \text{ अपरिमेय है,} \\ \tan(x) & \text{जब } x \text{ परिमेय है} \end{cases}$$

तब अंतराल $[0, 4\pi]$ में x के कितने मानों पर $f(x)$ सतत् होगा?

Ans. 4

Sol. The given function is defined using the two functions $x^2 + 1$ and $\tan(x)$. Both these functions are continuous wherever they are defined. Since every irrational number z has a non terminating, non repeating decimal expansion we see that given any $\epsilon > 0$ there is a rational number p such that the distance between z and p is less than ϵ . Using these facts one can see that the given function will be continuous precisely at those x in the interval $[0, 4\pi]$ where $x^2 + 1 = \tan(x)$. Since $x^2 + 1$ is positive, it will intersect $\tan(x)$ exactly once in the intervals $\left[0, \frac{\pi}{2}\right], \left[\pi, \frac{3\pi}{2}\right], \left[2\pi, \frac{5\pi}{2}\right], \left[3\pi, \frac{7\pi}{2}\right]$, as $\tan(x)$ increases from 0 to 1 in each of these intervals. $\tan(x)$ is negative elsewhere in the given domain. So we have 4 points of continuity.

D05/4/222. Let f be a continuous function from $\mathbb{R}$ to $\mathbb{R}$ (where $\mathbb{R}$ is the set of all real numbers) that satisfies the following property: For every natural number n , $f(n)$ = the smallest prime factor of n .

For example, $f(12) = 2$, $f(105) = 3$. Calculate the following.

(a) $\lim_{x \rightarrow \infty} f(x)$.

(b) The number of solutions to the equation $f(x) = 2016$.

माना $f : \mathbb{R} \rightarrow \mathbb{R}$ पर एक सतत् फलन है (जहाँ $\mathbb{R}$ सभी वास्तविक संख्याओं का समुच्चय है) जो निम्न गुणधर्म को संतुष्ट करता है : प्रत्येक प्राकृत संख्या n के लिए, $f(n)$, n का न्यूनतम अभाज्य गुणनफल है।

उदाहरण के लिए, $f(12) = 2$, $f(105) = 3$. निम्नलिखित का मान ज्ञात कीजिए।

(a) $\lim_{x \rightarrow \infty} f(x)$.

(b) समीकरण $f(x) = 2016$ के हलों की संख्या

Ans. (a) DNE (b) ∞

Sol. $f(x)$ will take value 2 for all even x . At the same time, primes provide an increasing infinite sequence of positive integers for which $f(x) = x$. Thus $\lim_{x \rightarrow \infty} f(x)$ does not exist. By intermediate value theorem, for each prime $p > 2016$ there is an x between p and $p + 1$ such that $f(x) = 2016$

D05/4/223. Find

$$\lambda = \max \{a \in \mathbb{Z} : a \leq \max(A)\}, \text{ where}$$

$$A = \{|12a + 4b + c| | ax^3 + bx^2 + cx + 5| \leq |e^x - e^2| \quad \forall x \geq 0\}$$

$$\lambda = \max \{a \in \mathbb{Z} : a \leq \max(A)\} \text{ का मान ज्ञात कीजिए, जहाँ}$$

$$A = \{|12a + 4b + c| | ax^3 + bx^2 + cx + 5| \leq |e^x - e^2| \quad \forall x \geq 0\}$$

Ans. 7

Sol.

D02/4/224. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function satisfying the relation $f(x+1) = \frac{1}{2} + \sqrt{f(x) - (f(x))^2}$ for all $x \in \mathbb{R}$. Then period of $f(x)$ is _____.

Ans. 2

Sol. Observe that $f(x) \geq 1/2$ for all $x \in \mathbb{R}$.

$$\text{Now } f(x+2) = f(x+1+1)$$

$$= \frac{1}{2} + \sqrt{f(x+1) - (f(x+1))^2}$$

$$= \frac{1}{2} + \sqrt{\frac{1}{2} + \sqrt{f(x) - (f(x))^2} - \frac{1}{4} - \sqrt{f(x) - (f(x))^2} - [f(x) - (f(x))^2]}$$

$$= \frac{1}{2} + \sqrt{\frac{1}{4} - f(x) + [f(x)]^2}$$

$$= \frac{1}{2} + \left| \frac{1}{2} - f(x) \right|$$

$$= \frac{1}{2} - \frac{1}{2} + f(x) \quad \left(\because f(x) \geq \frac{1}{2} \forall x \in \mathbb{R} \right)$$

$f(x)$

Therefore, 2 is period of $f(x)$.

D02/4/225. The real value of x satisfying $\sqrt[3]{20x + \sqrt[3]{20x + 13}} = 13$ can be expressed as $\frac{a}{b}$ where a and b are relatively prime positive integers. Find the value of b ? _____.

Ans. 5

Sol. $\sqrt[3]{20x + \sqrt[3]{20x + 13}} = 13$

$$\sqrt[3]{20x + \sqrt[3]{20x + \sqrt[3]{20x + \sqrt[3]{20x + \sqrt[3]{20x + \dots \infty}}}} = 13$$

$$\Rightarrow \sqrt[3]{20x + 13} = 13 \Rightarrow 20x = 133 - 13 \Rightarrow x = \frac{2184}{20} = \frac{546}{5}$$

D02/4/226. Find the number of real solutions to the equation $x = 99 \sin(\pi x)$. _____.

Ans. 199

Sol. The number of solutions is the number of times the line $y = \frac{x}{99}$ meets the graph of $y = \sin(\pi x)$. This can occur only for $x \in [-99, 99]$ because $\sin(\pi x)$ has range $[-1, 1]$. Also $\sin(\pi x)$ is periodic with period 2. For $x \geq 0$, the two graphs meet twice in each cycle of $\sin(\pi x)$, both intersections occurring in the first half of the cycle. There are 50 such half-cycles from $x = 0$ to $x = 99$, over intervals $[0, 1], [2, 3], \dots, [98, 99]$. So there are 100 non-negative solutions. Similarly there are 100 solutions ≤ 0 because both graphs are odd. Since $x = 0$ is counted twice, the total number of solutions is $100 + 100 - 1 = 199$.

D02/4/227. For sets A and B , let $f : A \rightarrow B$ and $g : B \rightarrow A$ be functions such that $f(g(x)) = x$ for each x . For each statement below, write whether it is TRUE or FALSE. _____.

(A) The function f must be one-to-one.

(B) The function f must be onto.

(C) The function g must be one-to-one.

(D) The function g must be onto

Ans. FTTF

Sol. If $g(x_1) = g(x_2)$, then $x_1 = f(g(x_1)) = f(g(x_2)) = x_2$, so g is one-to-one. Also f is onto because each $x \in B$ is in the image of f , namely $x = f(g(x))$. The other two statements are false, e.g. by constructing an example in which A is a larger finite set than B .

D02/4/228. Suppose $f(x)$ is a function from $\mathbb{R}$ to $\mathbb{R}$ such that $f(f(x)) = f(x)^{2013}$. Show that there are infinitely many such functions, of which exactly three are polynomials. (Here $\mathbb{R}$ = the set of real numbers.) [O]

_____.

Ans.

Sol. If f is a polynomial, then we make two cases. (i) If $f(x) = a$ constant c , then the given condition is equivalent to $c = c^{2013}$, which happens precisely for three values of c , namely $c = 0, 1, -1$ (since we have $c(c^{2012} - 1) = 0$, so $c = 0$ or $c^{2012} = 1$). Thus there are three constant functions with the given property. (ii) If $f(x)$ is a non-constant polynomial, then consider its range set $A = \{f(x) \mid x \in \mathbb{R}\}$. Now for all $a \in A$, we have by the given property $f(a) = a^{2013}$. So the polynomial $f(x) - x^{2013}$ has all elements of A as its roots. Since there are infinitely many values in A (e.g. applying the intermediate value theorem because f is continuous), the polynomial $f(x) - x^{2013}$ has infinitely many roots and thus must be the zero polynomial, i.e., $f(x) = x^{2013}$ for all real number x .

Note: One can also deduce that the degree of f must be 0 or 2013 by equating the degrees of $f(f(x))$ and $f(x)^{2013}$. Then, in the non-constant case, it is possible to argue first that the leading coefficient is 1 and then that all other coefficients must be 0.

To find infinitely many function with the given property, define $f(0) = 0$, $f(1) = 1$ and $f(-1) = -1$. For every other real number x , arbitrarily define $f(x)$ to be 0, 1 or -1 . It is easy to see that any such function satisfies the given property. (Other answers are possible, e.g., more systematically, observe that $f(a) = a^{2013}$ for at least one real number a (e.g., any number in the range of f) and then this forces $f(x) = x^{2013}$ for all

$x \in S = \{a^{2013^i} \mid i = 0, 1, 2, \dots\}$. We use this as follows. Fix a real number a . Then define $f(x) = x^{2013}$ for all

$x \in S = \{a^{2013^i} \mid i = 0, 1, 2, \dots\}$. For all $x \notin S$, simply define $f(x) =$ any element of the set S , e.g., a itself will do.)

D02/4/229. Notice that the quadratic polynomial $p(x) = 1 + x + \frac{1}{2}x(x-1)$ satisfies $p(j) = 2^j$ for $j = 0, 1$ and 2 . A polynomial $q(x)$ of degree 7 satisfies $q(j) = 2^j$ for $j = 0, 1, 2, 3, 4, 5, 6, 7$. Find the value of $q(10)$.

_____.

Ans. 968

Sol. Looking at the hint, recall that $\sum_{i=0}^n \binom{n}{i} = 2^n$. So, taking $\binom{x}{i} := \frac{x(x-1)\dots(x-i+1)}{i!}$, the polynomial

$p(x) = \sum_{i=0}^7 \binom{x}{i}$ satisfies the given requirements (note that for an integer value of x that is $< i$, the value of

$\binom{x}{i}$ is 0). Therefore $p(10) = 2^{10} - \binom{10}{8} - \binom{10}{9} - \binom{10}{10} = 1024 - 45 - 10 - 1 = 968$. (Given $n+1$ distinct

numbers x_i , and a list of numbers y_i , there always exists a polynomial $p(x)$ of degree at most n satisfying $p(x_i) = y_i$. One can write such $p(x)$ explicitly by Lagrange interpolation but in this particular case a trick could be used. Such a polynomial $p(x)$ is also unique because the difference between two such polynomials would have degree at most n but the difference would also have all $n+1$ numbers x_i as its roots, forcing the difference to be the zero polynomial.)

D02/4/230. Let $N = \{1, 2, 3, \dots\}$ be the set of natural integers and let

$$f: N \times N \rightarrow N$$

be defined by $f(m, n) = (2m-1) \cdot 2^n$. Is f injective? Is f surjective? Give reasons. _____.

Ans. Injective but not surjective

Sol. f is injective: suppose $f(m_1, n_1) = f(m_2, n_2)$ i.e. $(2m_1-1)2^{n_1} = (2m_2-1)2^{n_2}$. Then we must have $2^{n_2} = 2^{n_1}$ because the remaining part is odd, hence $n_1 = n_2$. This gives $2m_1-1 = 2m_2-1$ which implies $m_1 = m_2$.
 f is not surjective: there is no pair (m, n) for which $f(m, n) = 1$.

D02/4/231. (a) If a polynomial $f(x)$ has roots $r_1, r_2, \dots, r_n$, what are the roots of the polynomial $f(x+1)$?
 (b) Let f be a polynomial of degree three. Set $g(x) = f(x-1)$. Assume none of the roots of f are -1 . What are the roots of the polynomial $x^3 g(1/x)$? _____.

Ans. (a) $r_i - 1$ where $i = 1, 2, 3, \dots, n$ (b) $\frac{1}{r_i + 1}$ where $i = 1, 2, 3, \dots, n$

Sol. $h(x) = f(x+1)$. Then $h(r-1) = f(r-1+1) = f(r) = 0$ for every root of f . So the roots of h are $r-1$, r a root of f .

The roots of $g(x)$ are $r+1$, where r is a root of f . Since -1 is not a root of f , none of the roots of g are zero. Let $h(x) = x^3 \cdot g(1/x)$. If r is a root of g which is non zero, $h(1/r) = (1/r)^3 g(r)$ is zero. So the roots of $h(x)$ are $1/(r+1)$ where r is a root of f .

D02/4/232. Recall that if h is a function from X to Y and g is a function from Y to Z then, $g \circ h$ is the function from X to Z such that $(g \circ h)(x) = g(h(x))$, for all $x \in X$.

Let S be the set of all functions f from $\{1, 2, 3, 4, 5, 6\}$ to $\{1, 2, 3, 4, 5, 6\}$ such that $f \circ f = f$.

(a) Compute the number of functions $f \in S$ whose range has three elements.

(b) What is the cardinality of S ? _____.

Ans. (a) 540 (b) 1057

Sol. If y is an element in the range of f then $f(y) = y$ since f is idempotent. Because if z maps to y , $y = f(z) = f(f(z)) = f(y)$.

Can pick 3 elements in the range of f in $\binom{6}{3}$ ways. Now these elements have to go to themselves under f since $f \circ f = f$. And the remaining 3 elements have to map to one of them. So the number of such functions is $3 * 3 * 3 = 27$. So the total number of functions with range of size 3 is $\binom{6}{3} * 27$.

D02/4/233. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function satisfying $f(10 - x) = f(x)$ and $f(2 - x) = f(2 + x) \forall x \in \mathbb{R}$. If $f(0) = \lambda$ (where $\lambda \in \mathbb{R}$). Then the minimum possible number of values of x satisfying $f(x) = \lambda$ in the interval $[0, 38]$, is _____.

Ans. 13

Sol.

D02/4/234. Let $f : A \rightarrow \left[-\frac{1}{2}, 0\right]; f^3(x) - 3xf(x) + x^3 + 1 = 0$.

If f is an invertible function, then number of solutions of

$$(f(f(x)))^2 = \frac{3}{2}f^{-1}(x) + 1$$

is equal to _____.

Ans. 1

Sol.

D02/4/235. Let $\lfloor x \rfloor = \max\{a \in \mathbb{Z} : a \leq x\}; \{x\} = x - \lfloor x \rfloor$,

$$A = \left\{ x \in \mathbb{R} : \frac{\lfloor x+1 \rfloor^2}{6} = \frac{x}{3} + \frac{\{x\}}{2} \right\},$$

$$B = \left\{ x \in \mathbb{R}^+ : \{x^2\} = \left\{ \frac{1}{x} \right\}, x^2 \in (2, 3) \right\}.$$

If $|X|$ denotes the cardinal number of set X , then $|A| + |B|$ is equal to _____.

Ans. 4

Sol.

D02/4/236. If $1 < x^2 < 9$, then number of solutions of the equation $\{ |x| \} = \left\{ \frac{1}{x^2} \right\}$ (where $\{.\}$ denote fractional part function) is _____.

Ans. 4

D02/4/237. Let $|X|$ denotes the cardinal number of set X ,

$$A = \left\{ x \in \mathbb{R} : \sqrt{ax^2 + bx} \in \mathbb{R} \right\},$$

$$B = \left\{ \sqrt{ax^2 + bx} \in \mathbb{R} : x \in \mathbb{R} \right\},$$

$$C = \{ a \in \mathbb{R} : A = B \text{ for at least one } b \in \mathbb{R}^+ \}.$$

Then absolute value of sum of elements of set C is equal to _____.

Ans. 4

Sol.

D03/4/238. If $\sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \pi$, $x, y, z \geq 0$, and

$$x^4 + y^4 + z^4 + 4x^2y^2z^2 = k(x^2y^2 + y^2z^2 + z^2x^2) \text{ then value of } k \text{ is } \underline{\hspace{2cm}}.$$

यदि $\sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \pi$, $x, y, z \geq 0$, तथा

$$x^4 + y^4 + z^4 + 4x^2y^2z^2 = k(x^2y^2 + y^2z^2 + z^2x^2) \text{ तो } k \text{ का मान होगा}$$

Ans. 2

Sol. We have,

$$\Rightarrow \sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \pi$$

$$\Rightarrow \sin^{-1} x + \sin^{-1} y = \pi - \sin^{-1} z$$

$$\Rightarrow \cos(\sin^{-1} x + \sin^{-1} y) = \cos(\pi - \sin^{-1} z)$$

$$\Rightarrow \cos(\sin^{-1} x) \cos(\sin^{-1} y) - \sin(\sin^{-1} x) \sin(\sin^{-1} y) = -\cos(\sin^{-1} z)$$

$$\Rightarrow \sqrt{1-x^2} \sqrt{1-y^2} - xy = -\sqrt{1-z^2}$$

$$\left\{ \cos(\sin^{-1} x) = \cos(\cos^{-1} \sqrt{1-x^2}) = \sqrt{1-x^2} \right\}$$

$$\begin{aligned}
&\Rightarrow \sqrt{1-x^2}\sqrt{1-y^2} = xy - \sqrt{1-z^2} \\
&\Rightarrow (1-x^2)(1-y^2) = x^2y^2 + 1 - z^2 - 2xy\sqrt{1-x^2} \quad \{\text{squaring both sides}\} \\
&\Rightarrow 1 - x^2 - y^2 + x^2y^2 = x^2y^2 + 1 - z^2 - 2xy\sqrt{1-x^2} \\
&\Rightarrow x^2 + y^2 - z^2 = 2xy\sqrt{1-x^2} \quad \Rightarrow (x^2 + y^2 - z^2)^2 = 4x^2y^2(1-x^2) \\
&\Rightarrow x^4 + y^4 + z^4 - 2x^2z^2 - 2y^2z^2 + 2x^2y^2 = 4x^2y^2 - 4x^2y^2z^2 \\
&\Rightarrow x^4 + y^4 + z^4 + 4x^2y^2z^2 = 2(x^2y^2 + y^2z^2 + z^2x^2)
\end{aligned}$$

D03/4/239. Number of integral roots of the equation $[\sin^{-1}x] + [\cos^{-1}x] = 1$ is/are ([.] is GIF) _____.

Ans. 3

Sol. $[\sin^{-1}x] + [\cos^{-1}x] = 1$

$$x = 0 \quad 0 + \left[\frac{\pi}{2}\right] = 1$$

$$x = 1 \quad \left[\frac{\pi}{2}\right] + 0 = 1$$

$$x = -1 \quad \left[-\frac{\pi}{2}\right] + [\pi] = 1$$

D03/4/240. Let $f(x, y) = \frac{x^3}{2} \operatorname{cosec}^2\left(\frac{1}{2} \tan^{-1} \frac{x}{y}\right) + \frac{y^3}{2} \sec^2\left(\frac{1}{2} \tan^{-1} \frac{y}{x}\right)$ where x and y are of same sign, then

Find the value of $\sum_{x=1}^{10} \sum_{y=1}^{10} \left\{ f(x, y) - (x+y)(x^2 + y^2) \right\}$.

Ans. 0

Sol.

D03/4/241. If range of the function $f(x) = \left(\cos^{-1} \frac{x}{2}\right)^2 + \pi \sin^{-1} \frac{x}{2} - \left(\sin^{-1} \frac{x}{2}\right)^2 + \frac{\pi^2}{12}(x^2 + 6x + 8)$ is $[a\pi^2, b\pi^2]$, then find the value of $2(a+b)$ _____.

Ans. 5

Sol.

$$\begin{aligned}
 f(x) &= \left(\cos^{-1} \frac{x}{2} \right)^2 + \pi \sin^{-1} \frac{x}{2} - \left(\sin^{-1} \frac{x}{2} \right)^2 + \frac{\pi^2}{12} (x^2 + 6x + 8) \\
 &= \left(\cos^{-1} \frac{x}{2} + \sin^{-1} \frac{x}{2} \right) \left(\cos^{-1} \frac{x}{2} - \sin^{-1} \frac{x}{2} \right) + \pi \sin^{-1} \frac{x}{2} + \frac{\pi^2}{12} (x^2 + 6x + 8) \\
 &= \frac{\pi}{2} \cos^{-1} \frac{x}{2} - \frac{\pi}{2} \sin^{-1} \frac{x}{2} + \pi \sin^{-1} \frac{x}{2} + \frac{\pi^2}{12} (x^2 + 6x + 8) \\
 &= \frac{\pi}{2} \left(\cos^{-1} \frac{x}{2} + \sin^{-1} \frac{x}{2} \right) + \frac{\pi^2}{12} (x^2 + 6x + 8) = \frac{\pi^2}{4} + \frac{\pi^2}{12} (x^2 + 6x + 8)
 \end{aligned}$$

Domain of $f(x)$ is $[-2, 2]$

$f(x)$ is increasing in $[-2, 2]$

$$\therefore \text{Range of } f(x) \text{ is } \left[\frac{\pi^2}{4}, \frac{9\pi^2}{4} \right] = [a\pi^2, b\pi^2] \Rightarrow 2(a+b) = 2\left(\frac{1}{4} + \frac{9}{4}\right) = 5$$

D04/4/242. $\lim_{n \rightarrow \infty} \left(\sqrt{n^2 + n + 1} - \left[\sqrt{n^2 + n + 1} \right] \right)$ ($n \in \mathbb{I}$) where $[]$ denotes the greatest integer function _____.

$\lim_{n \rightarrow \infty} \left(\sqrt{n^2 + n + 1} - \left[\sqrt{n^2 + n + 1} \right] \right)$ ($n \in \mathbb{I}$) जहाँ $[]$ महत्तम पूर्णांक फलन को दर्शाता है _____.

Ans. $\frac{1}{2}$

Sol. $n < \sqrt{n^2 + n + 1} < n + 1$

Hence $\left[\sqrt{n^2 + n + 1} \right] = n$

$$\therefore \ell = \lim_{n \rightarrow \infty} \left(\sqrt{n^2 + n + 1} \right) = \frac{1}{2}$$

D04/4/243. Let $S_n = 1 + 2 + \dots + n$ and $P_n = \frac{S_2}{S_2 - 1} \cdot \frac{S_3}{S_3 - 1} \cdot \frac{S_4}{S_4 - 1} \dots \frac{S_n}{S_n - 1}$, where $n \in \mathbb{N}$, ($n \geq 2$). Then $\lim_{n \rightarrow \infty} P_n =$ _____.

माना $S_n = 1 + 2 + \dots + n$ तथा $P_n = \frac{S_2}{S_2 - 1} \cdot \frac{S_3}{S_3 - 1} \cdot \frac{S_4}{S_4 - 1} \dots \frac{S_n}{S_n - 1}$, जहाँ $n \in \mathbb{N}$, ($n \geq 2$) है। तब $\lim_{n \rightarrow \infty} P_n =$ _____.

Ans. 3

Sol. $S_n = \frac{n(n+1)}{2}$ and $S_{n-1} = \frac{(n+2)(n-1)}{2}$

$$\therefore \frac{S_n}{S_{n-1}} = \frac{n(n+1)}{2} \cdot \frac{2}{(n+2)(n-1)} = \left(\frac{n}{n-1}\right) \left(\frac{n+1}{n+2}\right)$$

$$\therefore P_n = \left(\frac{2}{1} \cdot \frac{3}{2} \cdot \frac{4}{3} \cdot \frac{5}{4} \cdots \frac{n}{n-1}\right) \left(\frac{3}{4} \cdot \frac{4}{5} \cdot \frac{5}{6} \cdots \frac{n+1}{n+2}\right)$$

$$= \left(\frac{n}{1}\right) \left(\frac{3}{n+2}\right)$$

D04/4/244. (i) By the binomial theorem $(\sqrt{2} + 1)^{10} = \sum_{i=0}^{10} C_i (\sqrt{2})^i$, where C_i are appropriate constants. Write the value of i for which $C_i (\sqrt{2})^i$ is the largest among the 11 terms in this sum.

(ii) For every natural number n , let $(\sqrt{2} + 1)^n = p_n + \sqrt{2}q_n$, where p_n and q_n are integers. Calculate

$$\lim_{n \rightarrow \infty} \left(\frac{p_n}{q_n}\right)^{10} \text{ _____}.$$

(i) द्विपद प्रमेय के द्वारा $(\sqrt{2} + 1)^{10} = \sum_{i=0}^{10} C_i (\sqrt{2})^i$, जहाँ C_i उपयुक्त नियतांक है। i का मान ज्ञात कीजिए जिसके लिये $C_i (\sqrt{2})^i$, 11 पदों के योग में अधिकतम है।

(ii) प्रत्येक प्राकृत संख्या n के लिए, माना $(\sqrt{2} + 1)^n = p_n + \sqrt{2}q_n$ है, जहाँ p_n तथा q_n पूर्णांक हैं। $\lim_{n \rightarrow \infty} \left(\frac{p_n}{q_n}\right)^{10}$ का मान ज्ञात कीजिए _____.

Ans. (i) 6 (ii) 32

Sol. (i) $i = 6$. One way : simplify the ratio $\frac{C_{i+1}(\sqrt{2})^{i+1}}{C_i(\sqrt{2})^i}$ and see that this ratio is >1 till $i = 5$ and < 1 from $i = 6$ onwards.

(ii) 32. Using binomial expansion see that $(\sqrt{2} - 1)^n = \pm(p_n - \sqrt{2}q_n)$, where the sign depends on the parity

of n. As $n \rightarrow \infty$, $(\sqrt{2}-1)^n \rightarrow 0$ since $(\sqrt{2}-1) < 1$. Thus $(p_n - \sqrt{2}q_n) \rightarrow 0$ and so $\frac{p_n}{q_n} \rightarrow \sqrt{2}$.

D04/4/245. Answer the following questions :

(a) Evaluate $\lim_{x \rightarrow 0^+} (x^{x^x} - x^x)$

(b) Let $A = \frac{2\pi}{9}$, i.e. $A = 40$ degrees. Calculate the following

$$1 + \cos A + \cos 2A + \cos 4A + \cos 5A + \cos 7A + \cos 8A.$$

(c) Find the number of solutions to $e^x = \frac{x}{2017} + 1$. _____.

निम्न प्रश्नों के उत्तर दीजिए :

(a) $\lim_{x \rightarrow 0^+} (x^{x^x} - x^x)$ ज्ञात कीजिए।

(b) माना $A = \frac{2\pi}{9}$, i.e. $A = 40$ डिग्री हैं। निम्न का मान ज्ञात करो

$$1 + \cos A + \cos 2A + \cos 4A + \cos 5A + \cos 7A + \cos 8A.$$

(c) $e^x = \frac{x}{2017} + 1$ के हलों की संख्या ज्ञात करो। _____.

Ans. (a) -1, (b) 1, (c) 2

Sol. (a) First consider the limit $\lim_{x \rightarrow 0^+} x^x = \lim_{x \rightarrow 0^+} (e^{\log_x x})$

$$= \lim_{x \rightarrow 0^+} \left(e^{\frac{\log x}{\frac{1}{x}}} \right)$$

$$\text{Now consider the following limit } \lim_{x \rightarrow 0^+} \frac{\log x}{\frac{1}{x}} = \lim_{x \rightarrow 0} \frac{\frac{1}{x}}{-\frac{1}{x^2}}$$

$$= 0$$

substituting the value 0 from (2) in equation (1) we get that the limit is 1. Now

$$\lim_{x \rightarrow 0^+} (x^{x^x} - x^x) = \lim_{x \rightarrow 0^+} x^{x^x} - \lim_{x \rightarrow 0^+} x^x$$

$$\begin{aligned}
&= \lim_{x \rightarrow 0^+} x^{\lim_{x \rightarrow 0^+} x} - \lim_{x \rightarrow 0^+} x^x \\
&= 0 - 1 \\
&= -1
\end{aligned}$$

(b) There are many ways to arrive at the answer 1. Here are two approaches. Let S be the above sum. Then

$$\begin{aligned}
S &= 1 + \cos A + \cos 2A + \cos 4A + \cos(2\pi - 4A) + \cos(2\pi - 2A) + \cos(2\pi - A) \\
&= 1 + 2(\cos A + \cos 2A + \cos 4A) \\
&= 1 + 2 \left(2 \cos \left(\frac{3A}{2} \right) \cos \left(\frac{A}{2} \right) + \cos \left(\pi - \frac{A}{2} \right) \right) \\
&= 1 + 2 \left(2 \cos \left(\frac{\pi}{3} \right) \cos \left(\frac{A}{2} \right) - \cos \left(\frac{A}{2} \right) \right) \\
&= 1 + 2 \left(2 \times \frac{1}{2} \cos \left(\frac{A}{2} \right) - \cos \left(\frac{A}{2} \right) \right) \\
&= 1
\end{aligned}$$

Recall that $\cos nA$ is the real part of e^{inA} . Then

$$\begin{aligned}
S &= \sum_{n=0}^8 \cos nA - \sum_{n=1}^2 \cos(3nA) \\
&= \operatorname{Re} \left(\sum_{n=0}^8 e^{inA} - \sum_{n=1}^2 e^{in \frac{2\pi}{3}} \right) \\
&= \operatorname{Re}(0 - \omega - \omega^2) \\
&= 1
\end{aligned}$$

Here ω is a complex cube root of unity.

(c) First, note that $x = 0$ is clearly a solution. Let $f(x) = e^x - \frac{x}{2017} - 1$. Then $x_0 = -\log 2107$ is the only critical point of $f(x)$. For all $x < x_0$ we have $f'(x) < 0$. Since $f(x) \rightarrow \infty$ as $x \rightarrow -\infty$ and $f(x) \rightarrow -\infty$ as $x \rightarrow \infty$ there is only one solution in the interval $(-\infty, x_0)$. For all $x > x_0$ we have $f'(x) > 0$ (i.e., $e^x > \frac{1}{2017}$). Hence there is only one solution in the interval (x_0, ∞) . In total there are exactly two solutions.

D04/4/246. Let $a_0, a_1, \dots, a_{19} \in \mathbb{R}$ and $P(x) = x^{20} + \sum_{i=0}^{19} a_i x^i$, $x \in \mathbb{R}$

If $P(x) = P(-x)$ for all $x \in \mathbb{R}$, and $P(k) = k^2$, for $k = 0, 1, 2, \dots, 9$ then find $\lim_{x \rightarrow 0} \frac{P(x)}{\sin^2 x}$ _____.

माना $a_0, a_1, \dots, a_{19} \in \mathbb{R}$ तथा $P(x) = x^{20} + \sum_{i=0}^{19} a_i x^i, x \in \mathbb{R}$ हैं।

यदि $P(x) = P(-x) \forall x \in \mathbb{R}$, तथा $k = 0, 1, 2, \dots, 9$ के लिए $P(k) = k^2$ तब $\lim_{x \rightarrow 0} \frac{P(x)}{\sin^2 x}$ ज्ञात कीजिए _____.

Ans. -362879

Sol. The relation $P(x) = P(-x)$ implies that

$$x^{20} + \sum_{i=0}^{19} a_i x^i = x^{20} + \sum_{i=0}^{19} (-1)^i a_i x^i,$$

$$\sum_{i=0}^{19} (1 - (-1)^i) a_i x^i = 0$$

All coefficients a_i that correspond to odd indices i are equal to zero. So,

$$P(x) = x^{20} + \sum_{i=0}^9 a_{2i} x^{2i}$$

Consider polynomial $Q(x) = x^{10} + \sum_{i=0}^9 a_{2i} x^i - x$. By assumption,

$$Q(k^2) = k^{20} + \sum_{i=0}^9 a_{2i} k^{2i} - k^2 = P(k) - k^2 = 0, 0 \leq k \leq 9$$

Observe that $a_2 - 1$ is the coefficient near x in $Q(x)$. 10 roots of the polynomial Q are known :

$$Q(x) = x(x-1)(x-2^2) \dots (x-9^2)$$

In particular, $a_0 = 0$ and

$$a_2 - 1 = -\prod_{k=1}^9 k^2, a_2 = 1 - (9!)^2 = -362879$$

Then $P(x) = -362879x^2 + \sum_{i=3}^{19} a_i x^i + x^{20}$ and

$$\lim_{x \rightarrow 0} \frac{P(x)}{\sin^2 x} = \lim_{x \rightarrow 0} \frac{x^2}{\sin^2 x} \lim_{x \rightarrow 0} \frac{P(x)}{x^2} = -362879$$

D04/4/247. If the value of limit $\lim_{x \rightarrow 0} \frac{(1+x)^{\frac{1}{x}} - e + \frac{ex}{2}}{x^2} = \frac{Ae}{B}$, where A and B are co-prime natural numbers, then find the

value of $(A+B)$ _____.

यदि $\lim_{x \rightarrow 0} \frac{(1+x)^{\frac{1}{x}} - e + \frac{ex}{2}}{x^2} = \frac{Ae}{B}$ है, जहाँ A तथा B सह अभाज्य प्राकृत संख्याएँ हैं, तब (A + B) का मान ज्ञात कीजिए

_____.

Ans. 35

Sol. $\lim_{x \rightarrow 0} \frac{(1+x)^{\frac{1}{x}} - e + \frac{ex}{2}}{x^2}$

$$\lim_{x \rightarrow 0} \frac{e^{\frac{1}{x} \ln(1+x)} - e + \frac{ex}{2}}{x^2}$$

$$\lim_{x \rightarrow 0} \frac{e^{\frac{1}{x} \left(x - \frac{x^2}{2} + \dots \right)} - e + \frac{ex}{2}}{x^2}$$

$$\lim_{x \rightarrow 0} \frac{e^{\left(1 - \frac{x}{2} + \frac{x^2}{3} + \dots \right)} - e + \frac{ex}{2}}{x^2}$$

$$\lim_{x \rightarrow 0} \frac{e^{\left(1 - \frac{x}{2} + \frac{x^2}{3} + \dots \right)} \left(-\frac{1}{2} + \frac{2x}{3} + \dots \right) + \frac{e}{2}}{2x}$$

$$\lim_{x \rightarrow 0} \frac{e^{\left(1 + \frac{x}{2} + \dots \right)} \left(-\frac{1}{2} + \frac{2x}{3} + \dots \right)^2 + e \left(\frac{2}{3} + \dots \right)}{2}$$

$$\lim_{x \rightarrow 0} \frac{e^{\left(\frac{1}{4} \right)} + \frac{2e}{3}}{2} \Rightarrow \frac{\frac{e}{4} + \frac{2e}{3}}{2} = \frac{11e}{24}$$

$$A = 11, B = 24$$

D09/4/248. A conical vessel is to be prepared out of a circular sheet of gold of unit radius. If $\lambda\pi$ is the sectorial area which is used to form the sheet so that the vessel has maximum volume, then the integer nearest to value of $\sqrt{300\lambda}$ is equal to _____.

Ans. 3

Sol.

D09/4/249. Find $13(\max A) + 2(\min B)$, if

$$A = \left\{ \frac{1}{|x-4|+1} + \frac{1}{|x-8|+1} \mid x \in \mathbb{R} \right\} \text{ and } B = \left\{ \frac{b+1}{a+b-2} \in \mathbb{R} \mid a^2 + b^2 = 1 \right\}.$$

Ans. 32

Sol.

D06/4/250. If P_n is the sum of a G.P. upto n terms ($n \geq 3$) such that $(1-r) \frac{dP_n}{dr} = aP_{n-1} + bP_n$, where r is common ratio of G.P. then maximum value of $|a+b|$ is _____.

यदि P_n , गु.श्रे. के n ($n \geq 3$) पदों का योग इस प्रकार है कि $(1-r) \frac{dP_n}{dr} = aP_{n-1} + bP_n$ है, जहाँ r गु.श्रे. का सार्वअनुपात है तब $|a+b|$ का अधिकतम मान होगा _____.

Ans. 1

Sol.
$$P_n = \frac{A(1-r^n)}{1-r}$$

$$\frac{dP_n}{dr} = A \left[\frac{(1-r)(-nr^{n-1}) + (1-r^n)}{(1-r)^2} \right]$$

$$(1-r) \frac{dP_n}{dr} = A \left[\frac{(1-r)(-nr^{n-1}) + (1-r^n)}{(1-r)} \right] = a \times A \left(\frac{1-r^{n-1}}{1-r} \right) + bA \left(\frac{1-r^n}{1-r} \right)$$

$$\Rightarrow -nr^{n-1} + nr^n + 1 - r^n = a - ar^{n-1} + b - br^n$$

$$\Rightarrow -nr^{n-1} + (n-1)r^n + 1 = -ar^{n-1} - br^n + a + b$$

$$a = n, b = 1 - n, a + b = 1$$

$$|a+b| = 1$$

D06/4/251. Let $p(x)$ be a polynomial of degree strictly less than 100 and such that it does not have $x^3 - x$ as a factor. If

$$\frac{d^{100}}{dx^{100}} \left(\frac{p(x)}{x^3 - x} \right) = \frac{f(x)}{g(x)}$$

for some polynomials $f(x)$ and $g(x)$ then find the smallest possible degree of $f(x)$. Here $\frac{d^{100}}{dx^{100}}$ means taking

the 100th derivative _____.

माना $p(x)$ ठीक 100 से कम घात का बहुपद है तथा $x^3 - x$ इसका गुणनखण्ड नहीं है। यदि $\frac{d^{100}}{dx^{100}} \left(\frac{p(x)}{x^3 - x} \right) = \frac{f(x)}{g(x)}$, किसी

बहुपद $f(x)$ तथा $g(x)$ के लिए है तब $f(x)$ की संभावित न्यूनतम घात होगी। यहाँ $\frac{d^{100}}{dx^{100}}$, 100वाँ अवकलज दर्शाता है _____.

Ans. 200

Sol. Using the division algorithm we have

$$\frac{p(x)}{x^3 - x} = q(x) + \frac{r(x)}{x^3 - x}$$

As the degree of $q(x)$ is strictly less than that of $p(x)$ its 100th derivative is certainly zero. As $x^3 - x$ is not a factor of $p(x)$ one may assume (without loss of generality) that $x^2 - 1$ divides $r(x)$. In that case we have

$$\begin{aligned} \frac{d^{100}}{dx^{100}} \left(\frac{p(x)}{x^3 - x} \right) &= \frac{d^{100}}{dx^{100}} \left(\frac{k}{x} \right) \\ &= \frac{100!k}{x^{100}} \end{aligned}$$

Hence the least possible degree of $f(x)$ is 0.

If one assumes that $x^3 - x$ doesn't divide $p(x)$ then we have

$$\frac{r(x)}{x^3 - x} = \frac{A'}{x} + \frac{B'}{x-1} + \frac{C'}{x+1}.$$

Consequently,

$$\frac{f(x)}{g(x)} = \frac{A}{x^{101}} + \frac{B}{(x-1)^{101}} + \frac{C}{(x+1)^{101}}$$

$$\therefore f(x) = A(x^2 - 1)^{101} + B(x^2 + x)^{101} + C(x^2 - x)^{101}$$

$$= (A + B + C)x^{202} + 101(B - C)x^{201} + \left(\binom{101}{2}B + \binom{101}{2}C - 101A \right)x^{200} + \dots$$

Choosing $B = C$ and $A + B + C = 0$ we see that the coefficient of x^{200} is $(101)(102) \neq 0$. Hence the least possible degree of $f(x)$ in this case is 200.

D08/4/252. A differentiable function $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfies $f(1) = 2$, $f(2) = 3$ and $f(3) = 1$. Show that $f'(x) = 0$ for some x .

_____.

Ans.

Sol. Since f is differentiable, it is continuous. By the intermediate value theorem, there is $a \in (2, 3)$ with $f(a) = 2 = f(1)$. Now by Rolle's theorem there is $x \in (1, a)$ with $f'(x) = 0$. –OR– The continuous function f over the closed interval $[1, 2]$ must attain its absolute maximum, which cannot be at either endpoint (since $f(2) > f(1)$ and $f(2) > f(3)$). So the maximum must be at an interior point x and then $f'(x) = 0$. –OR– By the mean value theorem, $f'(y) = 1 > 0$ for some $y \in (1, 2)$ and $f'(z) = -2 < 0$ for some $z \in (2, 3)$. So $f'(x) = 0$ for some $x \in (y, z)$ since for a differentiable f , the function f' satisfies the intermediate value property by Darboux's theorem. (This is important to say because f' need not be continuous.)

D08/4/253. Let $f(x) = \frac{x^4}{(x-1)(x-2)\dots(x-n)}$ where the denominator is a product of n factors, n being a positive integer. It is also given that the X-axis is a horizontal asymptote for the graph of f . Answer the independent questions below by choosing the correct option from the given ones.

- a. How many vertical asymptotes does the graph of f have?
(A) n (B) less than n (C) more than n (D) impossible to decide
- b. What can you deduce about the value of n ?
(A) $n < 4$ (B) $n = 4$ (C) $n > 4$ (D) impossible to decide
- c. As one travels along the graph of f from left to right, at which of the following points is the sign of $f(x)$ guaranteed to change from positive to negative?
(A) $x = 0$ (B) $x = 1$ (C) $x = n - 1$ (D) $x = n$
- d. How many inflection points does the graph of f have in the region $x < 0$?
(A) none (B) 1 (C) more than 1 (D) impossible to decide

Ans. (a) A (b) C (c) C (d) C

Sol. (a) n , at $x = 1, 2, \dots, n$.
 (b) $n > 4$, because $\lim_{x \rightarrow \pm\infty} f(x) = 0$ and for this to happen, the degree of the denominator of $f(x)$ must be greater than that of the numerator.
 (c) $x = n - 1$, because $f(x)$ is positive for $x > n$ and $f(x)$ changes sign precisely when it passes through $x = 1, 2, \dots, n$. Note that the sign of $f(x)$ for $x < 0$ and for $x \in (0, 1)$ depends on the parity of n .
 (d) more than 1. Note that $f(x) = 0$ only at $x = 0$, with multiplicity 4. Without loss of generality, let n be even. (If n is odd, the reasoning is completely parallel, see note at the end.) Now $f(x) > 0$ for $x < 1$ except at $x = 0$ and f has all derivatives for $x < 1$. Due to the multiple root at $x = 0$, the graph of f must be concave up (i.e. $f''(x) > 0$) near $x = 0$. Further, as $x \rightarrow -\infty$, the values of $f(x)$ stay positive and $\rightarrow 0$. Therefore, as one traces the graph leftward from the origin, it must become concave down at least once and eventually concave up again so as to approach the X-axis from above. (Note: If n is odd, $f(x) < 0$ for $x < 1$ except at $x = 0$. As one traces the graph leftward from the origin, the function is initially as well as eventually concave down and must be concave up at least once in-between so as to approach the X-axis from below.)

D08/4/254. Consider the function $f(x) = ax + \frac{1}{x+1}$, where a is a positive constant. Let L = the largest value of $f(x)$ and

S = the smallest value of $f(x)$ for $x \in [0, 1]$. Show that $L - S > \frac{1}{12}$ for any $a > 0$.

Ans.

Sol. Let $f(x) = ax + \frac{1}{x+1}$, We wish to understand the minimum and maximum of this function in the interval $[0, 1]$. Now $f(0) = 1$, $f(1) = a + \frac{1}{2}$ and $f'(x) = a - \frac{1}{(x+1)^2}$. Over the interval $[0, 1]$ the value of $f'(x)$

increases from $a - 1$ at $x = 0$ to $a - \frac{1}{4}$ at $x = 1$.

We should consider what happens to the sign of $f'(x)$. For this we consider the following cases.

(1) Suppose $a \leq 1/4$. Because $1/(x+1)^2 \geq 1/4$ on the interval $[0, 1]$, $f'(x) \leq 0$, so the maximum is at 0 and the minimum is at $x = 1$. So the difference is $1 - (1/2 + a) = 1/2 - a \geq 1/4 \geq 1/12$

(2) Suppose $a \geq 1$. Then $f'(x) \geq 0$ on the interval $[0, 1]$, so maximum is at 1 and minimum at 0. We get $a + 1/2 - 1 = a - 1/2 \geq 1/2 \geq 1/12$.

(3) Suppose $1/4 \leq a \leq 1$. Now $f'(x) = 0$ at $\tilde{x} = \frac{1}{\sqrt{a}} - 1$. For this range of a , $\tilde{x} \in [0, 1]$. In the interval

$[0, \tilde{x}]$, $f'(x) \leq 0$ and in the interval $[\tilde{x}, 1]$, $f'(x) \geq 0$. Now we make two sub-cases depending on at which endpoint the maximum occurs.

(3i) Suppose $1/4 \leq a \leq 1/2$. Then $f(0) \geq f(1)$. So minimum is at $\tilde{x}$, maximum is at $x = 0$. $f(\tilde{x}) = \sqrt{a} - a + \sqrt{a} = 2\sqrt{a} - a$. So the difference between maximum and minimum is $1 + a - 2\sqrt{a} = (1 - \sqrt{a})^2$. This is smallest when $\sqrt{a}$ is closest to 1 and so $(1 - \sqrt{a})^2 \geq (1 - 1/\sqrt{2})^2 = 3/2 - \sqrt{2}$. This is bigger than $1/12$ since $\left(\frac{3}{2} - \frac{1}{12}\right) = 17/12$ and $17^2 = 289 \geq 2 \times 12^2$.

(3ii) Suppose $1/2 \leq a \leq 1$. Now $f(1) \geq f(0)$. Max is at 1 and minimum is at $\tilde{x}$. The difference is $a + 1/2 - \sqrt{a} + a - \sqrt{a} = 2a - 2\sqrt{a} + 1/2 = \left(\sqrt{2a} - \frac{1}{\sqrt{2}}\right)^2$. By a calculation similar to the above it is bigger than $1/12$

D08/4/255. Find a polynomial $p(x)$ that simultaneously has both the following properties.

(i) When $p(x)$ is divided by x^{100} the remainder is the constant polynomial 1.

- (ii) When $p(x)$ is divided by $(x-2)^3$ the remainder is the constant polynomial 2.

Ans.

$$p(x) = A \left(\frac{x^{102}}{102} - \frac{4x^{101}}{101} + \frac{4x^{100}}{100} \right) + 1, \text{ Get A by putting } p(2) = 2$$

Sol.

Suppose a polynomial $f(x)$ leaves a constant remainder r when divided by the polynomial $(x-c)^k$. Then $f'(x)$ is divisible by $(x-c)^{k-1}$. The converse is also true: suppose for a polynomial $f(x)$, the derivative $f'(x)$ is divisible by $(x-c)^{k-1}$, say $f'(x) = q(x)(x-c)^{k-1}$. Then $f(x)$ leaves a constant remainder when divided by $(x-c)^k$. One can see this e.g. by substituting $u = (x-c)$ in $q(x)(x-c)^{k-1}$ and integrating.

In the given problem $p'(x)$ must be divisible by x^{99} as well as by $(x-2)^2$. Moreover any polynomial whose derivative is divisible by $x^{99}(x-2)^2$ will leave constant remainders when divided by either of x^{100} and $(x-2)^3$. The simplest way to find one such $p(x)$ is to integrate $Ax^{99}(x-2)^2 = A(x^{101} - 4x^{100} + 4x^{99})$ to get

$$p(x) = A \left(\frac{x^{102}}{102} - \frac{4x^{101}}{101} + \frac{4x^{100}}{100} \right) + B \text{ and solve for constants A and B to ensure desired values of the}$$

constant remainders. We have $p(0)=B=1$ and $p(2) = A \left(\frac{2^{102}}{102} - \frac{4 \times 2^{101}}{101} + \frac{4 \times 2^{100}}{100} \right) + 1 = 2$, which gives A.

Theoretical approach. Working through the following reasoning will be very useful for your understanding of basic arithmetic/algebra. It explains how to implement the Chinese remainder theorem using the Euclidean algorithm for finding GCD. This theorem states the following. One can always find an integer that leaves desired remainders when divided by two coprime integers a and b .

Suppose we are required to find an integer that leaves remainder r when divided by a and remainder s when divided by b . A way to achieve this systematically is to use the Euclidean algorithm, which finds GCD of two numbers by repeated division with remainder. This algorithm also enables one to write the GCD in the form $xa + yb$, where the integers x, y can be found explicitly by backward substitution in the steps used to calculate the GCD. If a and b are coprime, i.e. if their GCD is 1, then we can write $1 = xa + yb$. This tells you that xa is 1 modulo b and yb is 1 modulo a . Therefore, $sxa + ryb$ is r modulo a and s modulo b .

The relevance for this problem is that the same reasoning applies for polynomials in one variable, because in this setting too one has division with remainder. Because x^{100} and $(x-2)^3$ do not share a common factor, you know without any work that a polynomial with given properties must exist. The same algorithm as the previous paragraph (but now with polynomials) gives a systematic way to find it. In the given problem we could use a different trick because the specified remainders here were rather simple (constants). But there is a conceptual way as well by implementing the Chinese remainder theorem.

D08/4/256. A stationary point of a function f is a real number r such that $f'(r) = 0$. Consider a polynomial $p(x)$.

- (A) If $p(x)$ is of degree 2022, then $p(x)$ must have at least one stationary point.
- (B) If the number of distinct real roots of $p(x)$ is 2021, then $p(x)$ must have at least 2020 stationary points.
- (C) If the number of distinct real roots of $p(x)$ is 2021, then $p(x)$ can have at most 2020 stationary

points.

(D) If r is a stationary point of $p(x)$ AND $p''(r) = 0$, then the point $(r, p(r))$ is neither a local maximum nor a local minimum point on the graph of $p(x)$.

Ans. AB

Sol. (a) $p'(x)$ is a polynomial of degree 2021, which is odd, so it has a root by intermediate value theorem by looking at behaviour as $x \rightarrow \pm\infty$.
 (b) The graph of $p(x)$ has to turn between any two consecutive zeros, giving a stationary point, in fact a local max/min
 (c) The graph of $p(x)$ can turn more than once between zeros, or turn outside extreme zeros or have stationary points that are not maxima or minima.
 (d) is false, e.g., $p(x) = x^4$.

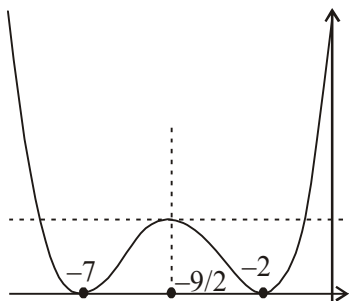
D08/4/257. Let a be a fixed real number. Consider the equation

$$(x+2)^2(x+7)^2 + a = 0, x \in \mathbb{R}$$

Where $\mathbb{R}$ is the set of real numbers. For what values of a , will the equation have exactly one double root?
 _____.

Ans. $-(5/2)^4$

Sol. Let $g(x) = (x+2)^2(x+7)^2$:



$$g'(x) = 2(x+2)(x+7)^2 + 2(x+2)^2(x+7) = 2(x+2)(x+7)(2x+9)$$

$g'(x) = 0$ if and only if $x \in \{-7, -9/2, -2\}$. Values $x = -7$ and $x = -2$ are roots of the equation $g(x) = 0$. This equation has two double roots. Value $x = -9/2$ is the root of the equation

$$g(x) = g(-9/2) = (5/2)^4$$

and this is the only double root of this equation.

Answer : $a = -(5/2)^4$

D08/4/258. Find λ , if

$$\left\{a \in \mathbb{R} \mid \left\{x \in \mathbb{R} \mid 3 - x^2 > |x - a|\right\} \text{ is NOT a NULL set} \right\} = \left(-\frac{13}{4}, \lambda\right)$$

Ans. 3

Sol.

D08/4/259. Let function f be defined as

$$f: \mathbb{R} \rightarrow \mathbb{R}; f(x) = ax + \cos 2x + \sin x + \cos x.$$

Further $m, n \in \mathbb{Z}$ such that

$$\{a \in \mathbb{R} \mid f \text{ is a strictly increasing function.}\} = \left[\frac{m}{n}, \infty\right); \gcd(m, n) = 1.$$

Then the value of $(m - n)$ is equal to

Ans. 9

Sol.

D08/4/260. Let p be a 6-degree polynomial satisfying

$$p(x) = p(2 - x) \quad \forall x \in \mathbb{R}, \quad \text{and} \quad |\{x \in \mathbb{R} \mid p(x) = 0\}| = 5.$$

If S denotes the sum of roots of polynomial p , and a is a positive constant satisfying

$$\{x \in \mathbb{R} \mid f''(x) > 0\} = \left(-\frac{\lambda}{a}, \frac{a}{\mu}\right), \quad f(x) = \begin{cases} xe^{ax}, & x \leq 0 \\ x + ax^2 - x^3, & x > 0 \end{cases}$$

then $\lambda + \mu + S$ is equal to (Here $|X|$ denotes the cardinal number of set X).

Ans. 11

Sol.

D07/4/261. Water is flowing in a container at a rate proportional to the square of the amount of water present inside the container. If at $t = 0$, volume of water inside the tank is V_0 & it takes 6 hours for the volume to become $2V_0$, then the time (in hours) taken for the volume to become $4V_0$ is equal to _____.

Ans. 9

Sol.

$$\text{Let } \frac{dV}{dt} = kV^2$$

$$\Rightarrow \int \frac{dV}{V^2} = \int k dt \Rightarrow -\frac{1}{V} = kt + C$$

$$\text{at } t = 0, V = V_0$$

$$\Rightarrow C = -\frac{1}{V_0} \Rightarrow \frac{1}{v} = \frac{1}{V_0} - kt$$

$$\text{at } t = 6, V = 2V_0$$

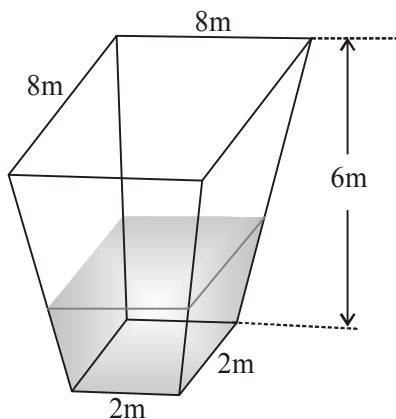
$$\Rightarrow 6k = \frac{1}{2V_0} \Rightarrow k = \frac{1}{12V_0}$$

$$\text{when } V = 4V_0$$

$$\Rightarrow kt = \frac{3}{4V_0} \Rightarrow \frac{3 \times 12V_0}{4V_0 \times 1}$$

$$t = 9 \text{ hours}$$

- D07/4/262. A pond has been dug at the IIT Bombay as an inverted truncated pyramid with a square base (see figure below). The depth of the pond is 6 m. The square at the bottom has side length 2 m and the top square has side length 8 m. Water is filled in at a rate of $\frac{19}{3}$ cubic meters per hour. At what rate is the water level rising exactly 1 hour after the water started to fill the pond ?

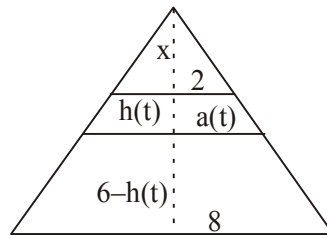


Ans. 19/27 m/h

Sol. Denote by $V(t)$ the volume of the water after t hours the water started to fill the pond. Then $V(t) = \frac{19}{3}t$

(cubic meters). Let $h(t)$ be the level of the water and $a(t)$ be the side of the square at the top of the water after t hours the water started to fill the pond.

Extend the truncated pyramid to get the non-truncated pyramid and let x be the height of the added part. Then we have the following picture



Similarity of triangles implies relations

$$\frac{x}{x + h(t)} = \frac{2}{a(t)}$$

$$\frac{x + h(t)}{x + 6} = \frac{a(t)}{8}.$$

Multiplying them we find

$$\frac{x}{x + 6} = \frac{1}{4}, x = 2.$$

Hence,

$$a(t) = h(t) + 2$$

Now,

$$\begin{aligned} V(t) &= \frac{h(t)}{3} (4 + a(t)^2 + 2a(t)) = \frac{h(t)}{3} (4 + h(t)^2 + 4h(t) + 4 + 2h(t) + 4) = \\ &= \frac{h(t)}{3} (h(t)^2 + 6h(t) + 12) = \frac{1}{3} ((h(t) + 2)^3 - 8) \end{aligned}$$

Differentiate this :

$$V'(t) = (h(t) + 2)^2 h'(t)$$

Then

$$h'(1) = \frac{V'(1)}{(h(1) + 2)^2} = \frac{V'(1)}{(3V(1) + 8)^{\frac{2}{3}}} = \frac{\frac{19}{3}}{9} = \frac{19}{27} \text{ (meters per second)}$$

D07/4/263. Let $P(x_1, y_1)$ be a point on curve

$$C : (x^2 - 11)(y + 1) + 4 = 0,$$

where $x_1, y_1 \in \mathbb{N}$. If the area of the triangle formed by normal drawn to C at P and the coordinate axes is equal to

$$\frac{a}{b}; a, b \in \mathbb{N} \quad \text{and} \quad \lambda = \min(a - 6b),$$

then find the value of λ _____.

Ans. 3

Sol.

T03/4/264. In ΔABC , circumradius is 3 and inradius is 1.5 units. If the value of $a \cot^2 A + b^2 \cot^3 B + c^3 \cot^4 C$ is $m\sqrt{n}$ where m and n are prime numbers, then find the value of $(m + n)$ _____.

ΔABC में परित्रिज्या 3 है तथा अन्तः त्रिज्या 1.5 है। यदि $a \cot^2 A + b^2 \cot^3 B + c^3 \cot^4 C$ का मान $m\sqrt{n}$ है जहाँ m तथा n अभाज्य संख्याएँ हैं तो $(m + n)$ का मान _____ होगा।

Ans. 16

Sol. Given $R : r = 3 : 1.5 = 2 \Rightarrow \Delta ABC$ must be equilateral.

$$\text{So } a = b = c = 2R \sin \frac{\pi}{3} = R\sqrt{3} \text{ (By sine rule)}$$

$$\text{Now } a \cot^2 A + b^2 \cot^3 B + c^3 \cot^4 C = R\sqrt{3} \left(\frac{1}{\sqrt{3}} \right)^2 + (R\sqrt{3})^2 \left(\frac{1}{\sqrt{3}} \right)^3 + (R\sqrt{3})^3 \left(\frac{1}{\sqrt{3}} \right)^4$$

$$= \frac{R}{\sqrt{3}} + \frac{R^2}{\sqrt{3}} + \frac{R^3}{\sqrt{3}} = \frac{3 + 3^2 + 3^3}{\sqrt{3}} = \frac{39}{\sqrt{3}} = 13\sqrt{3} = m\sqrt{n}$$

$$\text{Hence } (m + n) = 16$$

T03/4/265. In ΔABC , if A, B, C are in AP & $\frac{a \cos B + b \cos A}{c \cos A + a \cos C} = \frac{1}{\sqrt{3}}$, then the value of $\frac{3(\sin^2 A + \sin^2 B + \sin^2 C)}{(\sin 2B + \sin 2C)^2}$ is _____.

ΔABC में यदि A, B, C समान्तर श्रेणी में हैं तथा $\frac{a \cos B + b \cos A}{c \cos A + a \cos C} = \frac{1}{\sqrt{3}}$ तो $\frac{3(\sin^2 A + \sin^2 B + \sin^2 C)}{(\sin 2B + \sin 2C)^2}$ का मान _____ होगा।

Ans. 2

Sol.
$$\frac{a \cos B + b \cos A}{c \cos A + a \cos C} = \frac{1}{\sqrt{3}}$$

$$\frac{c}{b} = \frac{1}{\sqrt{3}} \quad \text{_____ (1)}$$

$$\text{and } (A + C) + B = 180^\circ \Rightarrow B = 60^\circ$$

from equation (1)

$$\frac{\sin C}{\sin B} = \frac{1}{\sqrt{3}} \Rightarrow \sin C = \frac{1}{2} \Rightarrow C = 30^\circ$$

and $A = 90^\circ$

$$\frac{3\left(1 + \frac{3}{4} + \frac{1}{4}\right)}{\left(\frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2}\right)^2} = \frac{3(2)}{3} = 2$$

T03/4/266. In a right angled triangle ABC, right angled at A, the radius of inscribed circle is 2 and radius of escribed circle touching side BC is 15, then length of side BC is 'k' where sum of digits of k is _____.

A पर समकोण किसी समकोण त्रिभुज ABC में अन्तः वृत्त की त्रिज्या 2 तथा भुजा BC को स्पर्श करने वाले बाह्य वृत्त की त्रिज्या 15 हो, तो भुजा BC की लम्बाई के 'k' है जहाँ k के अंको का योगफल _____ होगा।

Ans. 4

Sol. $\frac{\Delta}{s} = 2$ and $\frac{\Delta}{s-a} = 15$

$$\Rightarrow \frac{s-a}{s} = \frac{2}{15} \Rightarrow s = \frac{15}{13}a$$

$$\Rightarrow \Delta = \frac{30}{13}a = \frac{1}{2}bc$$

$$\Rightarrow bc = \frac{60}{13}a \quad \text{_____ (1)}$$

$$\text{Also } b+c = 2s-a = \frac{30}{13}a - a = \frac{17}{13}a \quad \text{_____ (2)}$$

$$\text{Now } b^2 + c^2 = a^2 \Rightarrow (b+c)^2 - 2bc = a^2$$

$$\Rightarrow \frac{289}{169}a^2 - \frac{120}{13}a = a^2 \text{ gives } a = 13$$

MATCH THE COLUMN TYPE QUESTION(S)

D03/5/267. Match the following column :

Column-I

Column-II

$$(a) f(x) = \begin{cases} x \left(\left[\frac{1}{x} \right] + \left[\frac{2}{x} \right] + \dots + \left[\frac{8}{x} \right] \right), & x \neq 0 \\ 9k, & x = 0 \end{cases} \text{ . The value of } k \quad (p) \ 1$$

such that f is continuous at $x = 0$ is

($[.]$ denotes the greatest integer function)

$$(b) f(x) = \begin{cases} \left(1 + x e^{-\frac{1}{x^2}} \sin \frac{1}{x^4} \right)^{e^{\frac{1}{x^2}}}, & x \neq 0 \\ k, & x = 0 \end{cases} \text{ . The value of } k \text{ such that} \quad (q) \ 2$$

f is continuous at $x = 0$ is

$$(c) f: [0, \infty) \rightarrow \mathbb{R}; f(x) = \begin{cases} \left(2 \sin \sqrt{x} + \sqrt{x} \sin \frac{1}{x} \right)^x, & x > 0 \\ k, & x = 0 \end{cases} \quad (r) \ 3$$

The value of k such that f is continuous at $x = 0$ is

$$(d) f: (0, \pi) \rightarrow \mathbb{R}; f(x) = \begin{cases} \frac{1 - \sin x}{(\pi - 2x^2)}; & x \neq \frac{\pi}{2} \\ k; & x = \frac{\pi}{2} \end{cases} \text{ . The value of} \quad (s) \ 4$$

$8\sqrt{|k|}$ such that f is continuous at $x = \frac{\pi}{2}$ is

Choose the correct option :

निम्न को सुमेलित कीजिए :

Column-I

Column-II

$$(a) f(x) = \begin{cases} x \left(\left[\frac{1}{x} \right] + \left[\frac{2}{x} \right] + \dots + \left[\frac{8}{x} \right] \right), & x \neq 0 \\ 9k, & x = 0 \end{cases} \text{ . } k \text{ का मान होगा यदि} \quad (p) \ 1$$

$f, x = 0$ पर सतत् है

($[.]$ महत्तम पूर्णांक फलन को दर्शाता है)

$$(b) f(x) = \begin{cases} \left(1 + x e^{-\frac{1}{x^2}} \sin \frac{1}{x^4} \right)^{e^{\frac{1}{x^2}}}, & x \neq 0 \\ k, & x = 0 \end{cases} \text{ . } k \text{ का मान होगा यदि} \quad (q) \ 2$$

$f, x = 0$ पर सतत् है

$$(c) f: [0, \infty) \rightarrow \mathbb{R}; f(x) = \begin{cases} \left(2 \sin \sqrt{x} + \sqrt{x} \sin \frac{1}{x}\right)^x, & x > 0 \\ k, & x = 0 \end{cases} \quad (r) \quad 3$$

$k, x = 0$ पर सतत् है

$$(d) f: (0, \pi) \rightarrow \mathbb{R}; f(x) = \begin{cases} \frac{1 - \sin x}{(\pi - 2x^2)}; & x \neq \frac{\pi}{2} \\ k; & x = \frac{\pi}{2} \end{cases}. \quad (s) \quad 4$$

$8\sqrt{|k|}$ का मान यदि $f, x = \frac{\pi}{2}$ पर अवकलनीय है

सही विकल्प का चयन कीजिए :

(A) A-s; B-p; C-p; D-p

(B) A-p; B-q; C-r; D-s

(C) A-q; B-r; C-p; D-s

(D) A-r; B-p; C-q; D-s

Ans.

A

Sol.

(A) Use Sandwich Theorem

$$x \left(\frac{1+2+3+\dots+8}{x} - 8 \right) < \left[\frac{1}{x} \right] + \left[\frac{2}{x} \right] + \dots + \left[\frac{8}{x} \right] \leq x \left(\frac{1+2+3+\dots+8}{x} \right)$$

$$\text{Taking limits find that } \lim_{x \rightarrow 0} x \left(\left[\frac{1}{x} \right] + \left[\frac{2}{x} \right] + \dots + \left[\frac{8}{x} \right] \right) = 36$$

$$(C) u = \left(2 \sin \sqrt{x} + \sqrt{x} \sin \frac{1}{x} \right)^x$$

$$\ell nu = x \ell n \left(2 \sin \sqrt{x} + \sqrt{x} \sin \frac{1}{x} \right)$$

$$= x \ell n \left(\frac{\left(2 \sin \sqrt{x} + \sqrt{x} \sin \frac{1}{x} \right) \sqrt{x}}{\sqrt{x}} \right) = x \ell n \sqrt{x} + x \ell n \left(\frac{2 \sin \sqrt{x}}{\sqrt{x}} + \sin \frac{1}{x} \right) = x \ell n \sqrt{x} + x g(x)$$

g is bounded

$$\text{Then } \lim_{x \rightarrow 0} = \lim_{x \rightarrow 0} \ell n \sqrt{x} + \lim_{x \rightarrow 0} x g(x)$$

$$= 0 + 0$$

$$\therefore u = e^0 = 1$$

D03/5/268. Let $f(x) = x^2 + ax + b$, $\forall x \in \mathbb{R}$, $f(0) > 0$ & $f(x)$ has integral roots. Tangent at $\left(\frac{5}{2}, p\right)$ to $y = f(x)$ is parallel to x-axis and $g(x) = f(x + 1)$.

Column-I

Column-II

- | | |
|--|----------|
| (a) $(a + b)$ can be | (p) -1 |
| (b) Value of $[p]$ can be (where $[.]$ represents greatest integer function) | (q) 1 |
| (c) Number of points where $g(x)$ is non-differentiable can be | (r) 3 |
| (d) Number of points where $ g(x) $ is non differentiable can be | (s) -3 |
| | (t) 5 |

Choose the correct option :

माना $f(x) = x^2 + ax + b$, $\forall x \in \mathbb{R}$, $f(0) > 0$ तथा $f(x)$ के पूर्णांक मूल हैं। बिन्दु $\left(\frac{5}{2}, p\right)$ पर $y = f(x)$ की स्पर्शरेखा x-अक्ष के समानान्तर है तथा $g(x) = f(x + 1)$.

Column-I

Column-II

- | | |
|---|----------|
| (a) $(a + b)$ हो सकता है | (p) -1 |
| (b) $[p]$ का मान हो सकता है (जहाँ $[.]$ महत्तम पूर्णांक फलन है) | (q) 1 |
| (c) बिन्दुओं की संख्या जहाँ $g(x)$ अवकलनीय नहीं है | (r) 3 |
| (d) बिन्दुओं की संख्या जहाँ $ g(x) $ अवकलनीय नहीं है | (s) -3 |
| | (t) 5 |

सही विकल्प चुनिए :

- | | |
|------------------------------|--------------------------|
| (A) A-r,t; B-q,s; C-p; D-p,t | (B) A-s,t; B-q; C-s; D-p |
| (C) A-p,q; B-p,s; C-q; D-r,t | (D) A-p; B-q; C-r; D-t |

Ans. C

Sol. $f(x) = x^2 + ax + b$

$$f'(x) \Big|_{\left(\frac{5}{2}, p\right)} = 0 \Rightarrow 2x + a \Big|_{\left(\frac{5}{2}, p\right)} = 0 \Rightarrow 2 \times \frac{5}{2} + a = 0$$

$$a = -5 \quad \therefore \quad f(x) = x^2 - 5x + b$$

Let its roots are α & β , $\alpha, \beta \in \mathbb{I}$

$$f(0) > 0 \quad b > 0$$

$$\alpha + \beta = 5 \quad \alpha\beta = b > 0$$

Both α & β are positive (α, β) can be (1,4) or (2,3).

$$\therefore \quad b = 4 \text{ or } 6$$

$$f(x) = x^2 - 5x + 4 \quad \text{or} \quad f(x) = x^2 - 5x + 6$$

D03/5/269. Let $f(x) = \begin{cases} \frac{e^{(a+2)x} - 1}{x} & , x < 0 \\ c & , x = 0 \text{ is continuous function } \forall x \in \mathbb{R} : \\ \frac{\left(x + b\left(x^4\right)\right)^{\frac{1}{2}} - x^{\frac{1}{2}}}{b\left(\frac{7}{x^2}\right)} & , x > 0 \end{cases}$

Column-I

- (a) Complete set of value(s) of a is
 (b) Complete set of value(s) of b is
 (c) Complete set of value(s) of c is
 (d) Complete set of value(s) of $2ac$ is

Column-II

- (p) $\mathbb{R}^+$
 (q) $\left\{-\frac{3}{2}\right\}$
 (r) $\left\{\frac{1}{2}\right\}$
 (s) $\mathbb{R} - \{0\}$
 (t) $\left\{-\frac{3}{2}, \frac{1}{2}\right\}$

Choose the correct option :

माना $f(x) = \begin{cases} \frac{e^{(a+2)x} - 1}{x} & , x < 0 \\ c & , x = 0 \text{ सतत् फलन है } \forall x \in \mathbb{R} : \\ \frac{\left(x + b\left(x^4\right)\right)^{\frac{1}{2}} - x^{\frac{1}{2}}}{b\left(\frac{7}{x^2}\right)} & , x > 0 \end{cases}$

स्तम्भ-I

- (a) a के मानों का पूर्ण समुच्चय
- (b) b के मानों का पूर्ण समुच्चय
- (c) c के मानों का पूर्ण समुच्चय
- (d) 2ac के मानों का पूर्ण समुच्चय

स्तम्भ-II

- (p) R^+
- (q) $\left\{-\frac{3}{2}\right\}$
- (r) $\left\{\frac{1}{2}\right\}$
- (s) $R - \{0\}$
- (t) $\left\{-\frac{3}{2}, \frac{1}{2}\right\}$

सही विकल्प चुनिए :

- (A) A-s; B-r; C-p; D-q
- (C) A-t,s; B-r,t; C-p,q,s; D-r

- (B) A-p; B-q; C-s; D-t
- (D) A-q; B-p; C-r; D-q

Ans. D

Sol.

D03/5/270. Consider the following

$$f(x) = g(g(x)); \text{ where } g(x) = \begin{cases} 1-x, & x \in [0, 1] \\ x+2, & x \in (1, 2) \\ 4-x, & x \in [2, 4] \end{cases}$$

Now, Match the following columns for function f

Column-I

- (a) $x = 0$
- (b) $x = 1$
- (c) $x = 2$
- (d) $x = 3$

Column-II

- (p) continuous
- (q) dis-continuous
- (r) differentiable
- (s) non-differentiable

Choose the correct option :

निम्न पर विचार कीजिए

$$f(x) = g(g(x)); \text{ जहाँ } g(x) = \begin{cases} 1-x, & x \in [0, 1] \\ x+2, & x \in (1, 2) \\ 4-x, & x \in [2, 4] \end{cases}$$

फलन f के लिए निम्न स्तम्भों को सुमेलित कीजिए

स्तम्भ-I**स्तम्भ-II**

(a) $x = 0$

(b) $x = 1$

(c) $x = 2$

(d) $x = 3$

Choose the correct option :

(A) A-p; B-q; C-r; D-s

(C) A-pr; B-ps; C-qs; D-qs

(p) सतत्

(q) असतत्

(r) अवकलनीय

(s) अवकलनीय नहीं

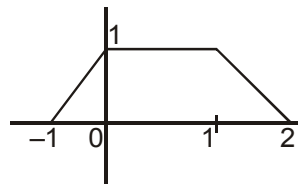
(B) A-qs; B-rs; C-pr; D-pq

(D) A-prs; B-rs; C-pqr; D-qrs

Ans. C

Sol.

D02/5/271. Given the graph of $y = f(x)$



List-I

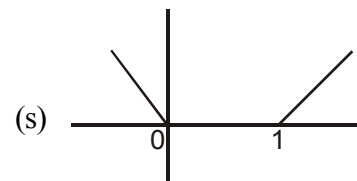
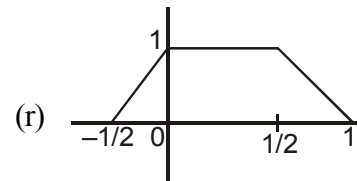
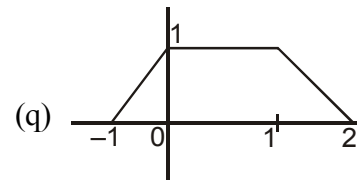
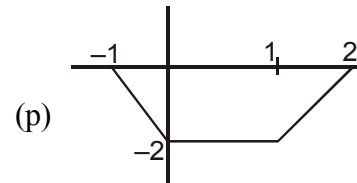
(a) $y = f(1 - x)$

(b) $y = f(2x)$

(c) $y = -2f(x)$

(d) $y = 1 - f(x)$

List-II



Codes :

(A) $a \rightarrow s$; $b \rightarrow q$; $c \rightarrow p$; $d \rightarrow r$

(B) $a \rightarrow s$; $b \rightarrow r$; $c \rightarrow p$; $d \rightarrow q$

(C) $a \rightarrow q; b \rightarrow r; c \rightarrow s; d \rightarrow p$

(D) $a \rightarrow q; b \rightarrow r; c \rightarrow p; d \rightarrow s$

Ans. D
Sol.

D02/5/272.

Column-I

Column-II

(a) $f(x) = \sin^2 2x - 2\sin^2 x$

(p) Range contains no natural number

(b) $f(x) = \frac{4}{\pi} (\sin^{-1}(\sin \pi x))$

(q) Range contains atleast one integer

(c) $f(x) = \sqrt{\ln(\cos(\sin x))}$

(r) Many one but not even function

(d) $f(x) = \tan^{-1} \left(\frac{x^2 + 1}{x^2 + \sqrt{3}} \right)$

(s) Both many one and even function

(t) Periodic but not odd function

Ans. A – P,Q,S,T ; B – Q, R ; C – P,Q,S ; D – P,S

Sol. (a) $f(x) = \sin^2 2x - 2 \sin^2 x$

$$= 2 \sin^2 x \cos 2x$$

Function is even, hence many one, function is also periodic

$$f(x) = (1 - \cos 2x) \cos 2x$$

$$= \frac{1}{4} - \left(\cos 2x - \frac{1}{2} \right)^2$$

$$\text{Range of function is } \left[-\frac{1}{4}, \frac{1}{4} \right]$$

(b) $f(x) = 4x$

(c) $f(x) = \sqrt{\ln(\cos(\sin x))}$

$$\ln(\cos(\sin x)) \geq 0$$

$$\Rightarrow \cos(\sin x) = 1$$

$$\Rightarrow f(x) = 0$$

(d) $f(x) = \tan^{-1} \left(\frac{x^2 + 1}{x^2 + \sqrt{3}} \right)$

$f(x)$ is even & hence many one

$$\text{Range is } \left[\frac{\pi}{6}, \frac{\pi}{4} \right)$$

D03/5/273. Match List-I with List-II and select the correct answer using the code given below the list.

Column-I

Column-II

$$(a) f: \mathbb{R} \rightarrow (-1, 1), f(x) = \frac{10^x - 10^{-x}}{10^x + 10^{-x}}$$

(p) one-one & onto

$$(b) f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = [x] + x$$

(q) many one & into

where $[x]$ denotes greatest integer function

$$(c) f: \mathbb{R} \rightarrow (0, \infty), f(x) = \begin{cases} e^x, & x < 0 \\ x^2 + \frac{1}{2}, & x \geq 0 \end{cases}$$

(r) one-one & into

$$(d) f: [-1, 1] \rightarrow \left[-\pi, \frac{\pi}{2}\right], f(x) = x \cos^{-1} x$$

(s) many one & onto

$$(A) a - p; b - r; c - q; d - s$$

$$(B) a - p; b - s; c - r; d - q$$

$$(C) a - r; b - q; c - s; d - p$$

$$(D) a - p; b - r; c - s; d - q$$

Ans.

D

Sol.

$$(P) f(x) = \frac{10^{2x} - 1}{10^{2x} + 1} = 1 - \frac{2}{10^{2x} + 1}$$

$$f'(x) > 0 \quad \forall x \in \mathbb{R}$$

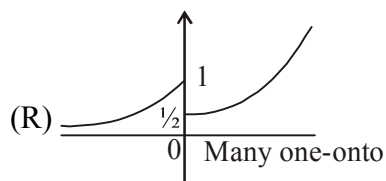
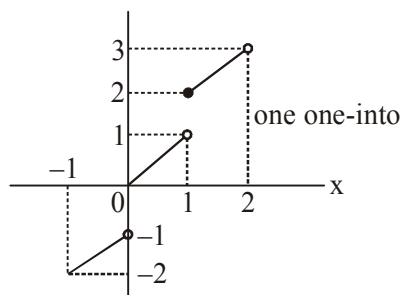
$\Rightarrow f(x)$ is increasing function

$\Rightarrow f(x)$ is one-one function

$$\text{Range of } f(x) = 1 - \frac{2}{(1, \infty)} \equiv 1 - 2(0, 1) \equiv (-1, 1)$$

$\Rightarrow f(x)$ is onto function

$$(Q) f(x) = [x] + x$$



$$(S) f(x) = x \cos^{-1} x$$

$$f'(x) = \frac{-x}{\sqrt{1-x^2}} + \cos^{-1} x$$

$$f'(1^-) \rightarrow -\infty$$

$$f'(-1^+) \rightarrow \infty$$

$f'(x)$ changes

its sign in its domain

$\Rightarrow f(x)$ is many one

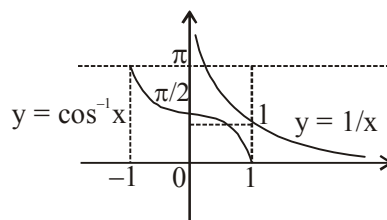
Let for $x \cos^{-1} x = 1$

$$\cos^{-1} x = \frac{1}{x} \text{ no solution}$$

$$\Rightarrow x \cos^{-1} x \neq 1$$

But codomain is $\left[-\pi, \frac{\pi}{2}\right]$

$\Rightarrow$ into function



D03/5/274. Match the following column :

Column-I

Column-II

(a) If $f(x) = \sin^{-1} x$ and $\lim_{x \rightarrow \frac{1}{2}^+} f(3x - 4x^3) = \ell - 3 \left(\lim_{x \rightarrow \frac{1}{2}^+} f(x) \right)$ (p) 3

then $[\ell]$ ($[\cdot]$ denotes greatest integer function)

(b) If $x > 1$ then the value of $\sin \left(\frac{1}{2} \tan^{-1} \frac{2x}{1-x^2} - \tan^{-1} x \right)$ is (q) -1

(c) Number of values of x satisfying $\sin^{-1} x - \cos^{-1} x = \sin^{-1} (3x - 2)$ (r) 2

(d) The value of $\sin \left(\tan^{-1} 3 + \tan^{-1} \frac{1}{3} \right)$ (s) 1

(A) $a \rightarrow p; b \rightarrow q; c \rightarrow r; d \rightarrow s$

(B) $a \rightarrow p; b \rightarrow s; c \rightarrow q; d \rightarrow r$

(C) $a \rightarrow q; b \rightarrow s; c \rightarrow p; d \rightarrow r$

(D) $a \rightarrow s; b \rightarrow r; c \rightarrow q; d \rightarrow p$

Ans.

A

Sol.

$$\sin^{-1} x - \cos^{-1} x = \sin^{-1} (3x - 2)$$

$$\text{domain is } x \in \left[\frac{1}{3}, 1 \right]$$

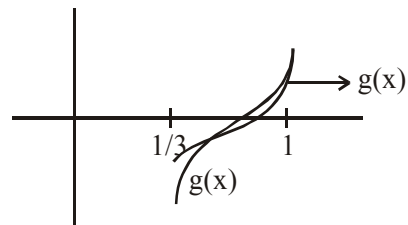
$$\text{Let } f(x) = \sin^{-1} x - \cos^{-1} x, g(x) = \sin^{-1} (3x - 2)$$

$$f'(x) = \frac{2}{\sqrt{1-x^2}}, \quad g'(x) = \frac{3}{\sqrt{1-(3x-2)^2}}$$

" $f(x)$ & $g(x)$ are increasing function

$$f(x) = 2 \sin^{-1} x - \frac{\pi}{2}$$

$$g(x) = \sin^{-1}(3x-2)$$



D09/5/275. Match the following column :

Column-I

Column-II

(a) Let $f(x) = x^3 + ax^2 + ax + 1$ has local extrema at $x = \alpha, \beta$

(p) $\frac{9}{2}$

where $\alpha < \beta$. If $f(\alpha) + f(\beta) = 2$ then the value of 'a' equals

(b) Value of $p + q$ for which $f(x) = x^3 + px^2 + qx + r$,

(q) $\frac{11}{2}$

where $p, q, r \in \mathbb{R}$ is monotonically decreasing in largest possible

interval $\left(-\frac{5}{3}, -1\right)$, is

(c) If the equation $e^{2x} = k\sqrt{x}$ has exactly 2 distinct solutions then 'k' can be equal to

(r) 7

(s) 9

Choose the correct option :

(A) a-s, b-q, c-p

(B) a-s, b-r, c-p, r, s

(C) a-p, b-s, c-q, r, s

(D) a-q, b-r, c-p

Ans.

C

Sol.

(A) Given $f(\alpha) + f(\beta) = 2$

where α, β are points of extrema of $f(x)$ and $\alpha < \beta$.

Using the symmetric nature of a cubic polynomial about its point of inflection, we transform $f(x)$ into :

$$f(x) = \left(x + \frac{a}{3}\right)^3 + ax + 1 - \frac{a^2x}{3} - \frac{a^3}{27} = \left(x + \frac{a}{3}\right)^3 + \left(a - \frac{a^2}{3}\right)\left(x + \frac{a}{3}\right) + 1 - \frac{a^2}{3} + \frac{2a^3}{27} \quad \dots(1)$$

Also, α, β satisfy $f'(\alpha) = f'(\beta) = 0$

$$f'(x) = 3\left(x + \frac{a}{3}\right)^2 + a - \frac{a^2}{3}$$

$$\Rightarrow f'(x) = 0 \text{ iff } \left(x + \frac{a}{3}\right)^2 = \frac{1}{3}\left(\frac{a^2}{3} - a\right)$$

$$\Rightarrow \alpha + \frac{a}{3} = -\frac{1}{3}\sqrt{a^2 - 3a}$$

$$\beta + \frac{a}{3} = \frac{1}{3}\sqrt{a^2 - 3a}$$

Using these values of $\alpha + \frac{a}{3}$ and $\beta + \frac{a}{3}$ in (i), we get :

$$f(\alpha) + f(\beta) = 2\left(1 - \frac{a^2}{3} + \frac{2a^3}{27}\right) = 2$$

$$\Rightarrow a = 0, \frac{9}{2}$$

$$a = 0 \text{ will make } \alpha = \beta \Rightarrow a = \frac{9}{2}$$

(B) Since the largest possible interval in which $f(x)$ decrease monotonically is $\left(-\frac{5}{3}, -1\right)$, graph of $f(x)$ is of

the kind

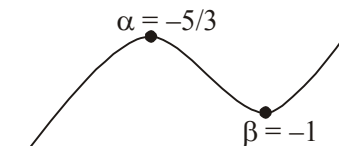
where α, β are roots of $f'(x) = 3x^2 + 2px + q$

$$\Rightarrow q = 5 \text{ and } p = 4$$

(C) Domain of $k\sqrt{x}$ is $[0, \infty)$

At $x = 0$, $e^{2x} = 1$ and range of e^{2x} is $[1, \infty)$

$$\text{Range of } k\sqrt{x} = \begin{cases} (-\infty, 0] & k \in (-\infty, 0) \\ \{0\} & k = 0 \\ [0, \infty) & k \in (0, \infty) \end{cases}$$



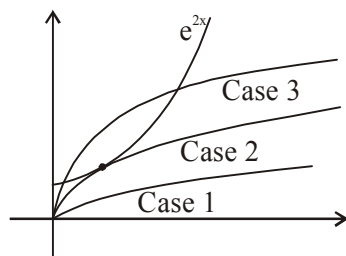
$e^{2x} = k\sqrt{x}$ has a solution only for $k \in (0, \infty)$

$$e^{2x} > k\sqrt{x} \text{ at } x = 0$$

Also, $x \rightarrow \infty$, $e^{2x} > k\sqrt{x}$ as exponential function will always dominate as x becomes large. e^{2x} and $k\sqrt{x}$ can be equal for some x only when k is large enough that $e^{2x} < k\sqrt{x}$ for some $c \in (0, \infty)$.

Then $e^{2x} = k\sqrt{x}$ will've distinct 2 distinct real roots : One each in $(0, c)$ and (c, ∞)

Let e^{2x} and $k\sqrt{x}$ touch each other for some $k = k_0 \in (0, \infty)$.



Case 1 : k small : $e^{2x} = k\sqrt{x}$ has no solution

Case 2 : $e^{2x} = k_0\sqrt{x}$ has one solution as the 2 graphs touch each other.

Case 3 : $e^{2x} = k\sqrt{x}$ has 2 solutions as the 2 graphs touch each other at 2 distinct points.

Graphically, we can see that 3 cases arise : $e^{2x} = k\sqrt{x}$ has 2 distinct solutions for $k \in (k_0, \infty)$

Since e^{2x} and $k_0\sqrt{x}$ touch each other, let the common point be α .

$$\Rightarrow e^{2\alpha} = k_0\sqrt{\alpha}$$

$$\text{Also } 2e^{2\alpha} = \frac{k_0}{2\sqrt{\alpha}} \quad \{\text{Tangents have equal slope at } x = \alpha\}$$

$$\Rightarrow 2k_0\sqrt{\alpha} = \frac{k_0}{2\sqrt{\alpha}} \Rightarrow \alpha = \frac{1}{4}$$

$$\Rightarrow k_0 = 2\sqrt{e}$$

Hence, $e^{2x} = k\sqrt{x}$ has 2 distinct solutions if $k \in (2\sqrt{e}, \infty)$

D09/5/276. Match the following column :

Column-I

Column-II

(a) $f(x) = \cos \pi x + 10x + 3x^2 + x^3$. The absolute minimum value of

(p) $\frac{3}{4}$

$f(x) \forall x \in [-2, 2]$ is

(b) If $x \in [-1, 1]$, then the minimum value of $f(x) = x^2 + x + 1$ is

(q) 2

(c) Let $f(x) = \frac{4}{3}x^3 - 4x$, $0 \leq x \leq 2$. Then, the global minimum value of the function

(r) -15

(d) Let $f(x) = 6 - 12x + 9x^2 - 2x^3$, $1 \leq x \leq 4$. Then, the absolute

(s) $-\frac{8}{3}$

maximum value of $f(x)$ in the interval is

Choose the correct option :

(A) a-r, b-p, c-s, d-q (B) a-p, b-p, c-r, d-s (C) a-q, b-p, c-r, d-q (D) a-q, b-p, c-r, d-p

Ans. A

Sol. P. $f'(x) = -\pi \sin \pi x + 10 + 6x + 3x^2 = 3(x+1)^2 + 7 - \pi \sin \pi x > 0$ for all x
 $f(x)$ is minimum in $-2 \leq x \leq 3$. So, the absolute minimum = $f(-2) = -15$

Q. $f'(x) = 2x + 1$. Therefore, for $-1 \leq x < -\frac{1}{2}$, we get

$f'(x) < 0$ and for $-\frac{1}{2} < x \leq 1$, we get $f'(x) > 0$

$f(x)$ is minimum decreases in $f\left[-1, -\frac{1}{2}\right]$

and minimum decreases in $\left(-\frac{1}{2}, 1\right]$.

$$\text{Min } f(x) = f\left(-\frac{1}{2}\right) = \left(-\frac{1}{2}\right)^2 + \left(-\frac{1}{2}\right) + 1 = \frac{3}{4}$$

R. $f'(x) = 4(x^2 - 1)$. So, for $0 \leq x < 1$, we get

$f'(x) < 0$, i.e., $f(x)$ is monotonically decreasing and for $1 < x \leq 2$, we get

$f'(x) > 0$, $f(x)$ is monotonically increasing.

$$\text{Min } f(x) = f(1) = \frac{4}{3} - 4 = -\frac{8}{3}$$

$$\text{S. } f'(x) = -12 + 18x - 6x^2 = -6(x^2 - 3x + 2) = -6(x - 1)(x - 2)$$

$f'(x) > 0$, if $1 < x < 2$ and $f'(x) < 0$, if $2 < x \leq 4$

$f(x)$ is monotonically increasing in $1 < x < 2$ and monotonically decreasing in $2 < x \leq 4$

Absolute maximum = the greatest among $\{f(1), f(2)\}$ = the greatest among $\{1, 2\} = 2$

D09/5/277. Match the following column :

Column-I

(a) If $x^2 + y^2 = 1$, then minimum value of $x + y$ is

(b) If maximum value of $y = a \cos x - \frac{1}{3} \cos 3x$ occurs at

$x = \frac{\pi}{6}$, then value of 'a' is

(c) If $f(x) = x - 2 \sin x$, $0 \leq x \leq 2\pi$ is increasing in the interval $(a\pi, b\pi)$, then $a + b$ is

(d) If equation of tangent to the curve $y = -e^{-\frac{x}{2}}$ where it crosses the y-axis is

$\frac{x}{p} + \frac{y}{q} = 1$, then $p - q$ is

Column-II

(p) -3

(q) $-\sqrt{2}$

(r) 3

(s) 2

Choose the correct option :

(A) a-s, b-p, c-q, d-r (B) a-q, b-s, c-s, d-r (C) a-p, b-r, c-q, d-s (D) a-p, b-r, c-s, d-q

Ans. B

Sol. (a) Let $x = \sin \theta$ and $y = \cos \theta$

$$x + y = \sin \theta + \cos \theta = \sqrt{2} \sin\left(\theta + \frac{\pi}{4}\right)$$

$$\text{Minimum value} = -\sqrt{2}$$

$$(b) y = a \cos x - \frac{1}{3} \cos 3x$$

$$y'\left(\frac{\pi}{6}\right) = 0$$

$$-a \sin\left(\frac{\pi}{6}\right) + \sin \frac{\pi}{2} = 0 \Rightarrow a = 2$$

$$(c) f'(x) = 1 - 2\cos x$$

$$f'(x) > 0 \Rightarrow \cos x < \frac{1}{2}$$

$$x \in \left(\frac{\pi}{3}, \frac{5\pi}{3}\right) \Rightarrow a + b = 2$$

$$(d) \text{ At } x = 0, y = -1 \text{ also } y'(0) = \frac{1}{2}$$

$$\text{Equation of tangent } y + 1 = \frac{1}{2}(x - 0)$$

$$\frac{x}{2} + \frac{y}{-1} = 1 \Rightarrow p - q = 3$$

D06/5/278.

Column-I

Column-II

$$(a) \text{ If } \int_0^{\pi} \frac{\log \sin x}{\cos^2 x} dx = -K \text{ then the value of } \frac{3k}{\pi} \text{ is greater than}$$

(p) 0

$$(b) \text{ If } e^{x+y} + e^{y-x} = 1 \text{ and } y'' - (y')^2 + K = 0, \text{ then } K \text{ is equal to}$$

(q) 1

$$(c) \text{ If } f(x) = x \ln x \text{ then } 2(f^{-1})'(\ln 4) \text{ is more than}$$

(r) 2

$$(d) \lim_{x \rightarrow \infty} (x \ln x)^{\frac{1}{x^2+1}} \text{ is less than}$$

(s) 4

(t) 5

Column-I

Column-II

$$(a) \text{ यदि } \int_0^{\pi} \frac{\log \sin x}{\cos^2 x} dx = -K \text{ तब } \frac{3k}{\pi} \text{ का मान किससे अधिक है}$$

(p) 0

$$(b) \text{ यदि } e^{x+y} + e^{y-x} = 1 \text{ तथा } y'' - (y')^2 + K = 0, \text{ तब } K \text{ का मान है}$$

(q) 1

$$(c) \text{ यदि } f(x) = x \ln x \text{ तब } 2(f^{-1})'(\ln 4) \text{ अधिक है}$$

(r) 2

$$(d) \lim_{x \rightarrow \infty} (x \ln x)^{\frac{1}{x^2+1}} \text{ कम है}$$

(s) 4

(t) 5

- (A) $a \rightarrow s, t; b \rightarrow p; c \rightarrow r, s, t; d \rightarrow p, q$
 (C) $a \rightarrow p, q, r; b \rightarrow q; c \rightarrow p, q; d \rightarrow r, s, t$

- (B) $a \rightarrow t; b \rightarrow s; c \rightarrow r; d \rightarrow q$
 (D) $a \rightarrow q, s, t; b \rightarrow p, r, t; c \rightarrow s, r; d \rightarrow p, q$

Ans.

C

Sol.

(A) (P, Q, R)

$$\text{let } I = \int_0^{\pi} \frac{\log(\sin x)}{\cos^2 x} dx$$

$$I = \int_0^{\pi/2} 2 \frac{\log(\sin x)}{\cos^2 x} dx = 2 \left[\int_0^{\pi/2} \log(\sin x) \sec^2 x dx \right] = 2 \left[\log(\sin x) \tan x \Big|_0^{\pi/2} - \int_0^{\pi/2} \frac{\cos x}{\sin x} \tan x dx \right]$$

$$= 2(0 - 0) - 2 \int_0^{\pi/2} dx = 0 - 2(\pi/2) = -\pi = -k \Rightarrow k = \pi$$

$$\therefore \frac{3k}{\pi} = 3 > 0, 1, 2$$

(B) (Q)

$$e^{x+y} + e^{y-x} = 1$$

$$e^x + e^{-x} = e^{-y}$$

$$e^x - e^{-x} = e^{-y}(-y')$$

$$e^x + e^{-x} = e^{-y}(-y'') + e^{-y}(y')^2$$

$$e^{-y} = e^{-y}(-y'') + e^{-y}(y')^2 \Rightarrow y' - (y')^2 + 1 = 0$$

$$\therefore k = 1$$

(C) (R, S, T)

$$\text{let } f^{-1} = g$$

$$f(f(x)) = x \Rightarrow (g'f(x)) f'(x) = 1$$

$$g'(2\ln 2) f'(2) = 1$$

$$g'(2\ln 2) = \frac{1}{1 + \ln 2}$$

$$2(f^{-1})'(\ln 4) = \frac{2}{1 + \ln 2} > 0, 1$$

(D) R, S, T

$$\ell = \lim_{x \rightarrow \infty} (x \ln x)^{1/x^2+1}$$

$$\ln \ell = \lim_{x \rightarrow \infty} \frac{\ell n x + \ln(\ell n x)}{x^2 + 1}$$

$$= \lim_{x \rightarrow \infty} \frac{1/x + 1/\ell n x \cdot 1/x}{2x} = 0$$

$$\ln(\ell) = 0 \Rightarrow \ell = 1$$

D08/5/279. Consider $f(x) = |\ln|x|| - kx^2$, $x \neq 0$. Match the column I with the value of k in column 2 :

Column-I

- (a) $f(x) = 0$ has two distinct solutions
 (b) $f(x) = 0$ has four distinct solutions
 (c) $f(x) = 0$ has six distinct solutions
 (d) $f(x) = 0$ has no solution

Column-II

- (p) $k = 0$
 (q) $k = \frac{1}{2e}$
 (r) $k \in \left(\frac{1}{2e}, \infty\right)$
 (s) $k \in (-\infty, 0)$
 (t) $k \in \left(0, \frac{1}{2e}\right)$

Choose the correct option :

(A) a-p,r; b-q, c-t, d-s

(B) a-q; b-p, c-s, d-t

(C) a-s,t; b-q,r, c-p,q,r, d-p,q,s

(D) a-t; b-s, c-q, d-r

Ans.

A

Sol.

$f(x) = |\ln|x|| - kx^2$ is an even function.

$= |\ln x| - kx^2$ when $x > 0$

Since $f(x)$ is even, for every $\alpha > 0$

This satisfies $f(\alpha) = 0$, $f(-\alpha) = 0$ as well.

Hence, to examine solution of $f(x) = 0$, we can restrict ourselves to $x \in (0, \infty)$ and all solutions lying in $(-\infty, 0)$ will be obtained automatically.

Number of solutions of $f(x) = 0$ is twice the number of points of intersection of the curves :

$y = |\ln x|$ and $y = kx^2$

The following cases arise : (Graphically or otherwise)

Case 1 : $k < 0$

Range of $|\ln x|$ is $[0, \infty)$

Range of kx^2 is $(-\infty, 0]$

No point of intersection as $|\ln x| = 0$ at $x = 1$ and $kx^2 = 0$ at $x = 0$

Case 2 : $k = 0$

1 point of intersection $\Rightarrow$ 2 solutions of $f(x) = 0$; $x = \pm 1$

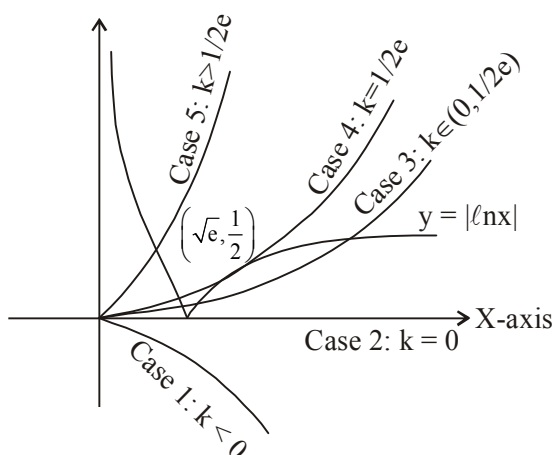
Case 3 : $k \in (0, k_0)$ where k_0 is the value of k such that $k_0 x^2$ and $|\ln x|$ touch each other for some $x \in (1, \infty)$.

3 points of intersection : 1 in $(0, 1)$ and 2 in $(1, \infty) \Rightarrow$ 6 solutions of $f(x) = 0$

Case 4 : $k = k_0$

$$\Rightarrow \ln x = k_0 x^2 \text{ and } \frac{1}{x} = 2k_0 x \text{ for some } x = \alpha \in (1, \infty)$$

$$\Rightarrow \ln \alpha = k_0 \alpha^2 = \frac{1}{2} \Rightarrow \alpha = \sqrt{e} \text{ and } k_0 = \frac{1}{2e}$$



Then 4 solutions of $f(X) = 0$ because 2 points of intersection : 1 in $(0,1)$ and $\alpha \in (1, \infty)$

Case 5 : $k > k_0 = \frac{1}{2e}$

1 point of intersection in $(0,1)$ and $\alpha \in (1, \infty)$

Note : In $(0,1)$, $|\ln x| = -\ln x \in (0, \infty)$ and $kx^2 \in (0, \infty)$ and $kx^2 \in (0, k)$

Also $|\ln x|$ decreases from $\infty \rightarrow 0$ and kx^2 increases from $0 \rightarrow k$

$\Rightarrow$ Exactly 1 point of intersection in $(0,1) \Rightarrow 2$ solutions for $f(X) = 0$

Consolidating the analysis from 5 cases :

$f(x) = 0$ has 0 solutions for $k \in (-\infty, 0)$

2 solutions for $k \in \{0\} \cup \left(\frac{1}{2e}, \infty\right)$

4 solutions for $k \in \frac{1}{2e}$

6 solutions for $k \in \left(0, \frac{1}{2e}\right)$

D08/5/280. Let f be a polynomial of degree 4 having real coefficients satisfying

$$f'(0) = f'(1) = f'(-1) = 0 \text{ and } f(0) = 4, f''\left(\frac{1}{2}\right) = -1$$

Match the following column :

Column-I

- (a) $f(x) = 0$ has
- (b) $4 - f(x) = 0$ has
- (c) $f'(x) + x - 1 = 0$ has

Column-II

- (p) root at $x = 2$
- (q) root at $x = 1$
- (r) 2 equal real roots

(d) $x f'(x) - 4 f(x) = 0$ has

(s) no real roots

Choose the correct option :

(A) a-s; b-r; c-q, r; d-p

(B) a-p; b-q; c-p; d-s

(C) a-p, r, s; b-q, s; c-p; d-r

(D) a-q; b-p; c-s; d-r

Ans. A

Sol. Using the conditions, we get $f(x) = x^4 - 2x^2 + 4$

D07/5/281. Match the curves in Column-I with curves in Column-II, if they touch each other.

Column-I

Column-II

(a) $y^2 = 4x$

(p) $x^2 + 676y^2 = 676$

(b) $xy = 4$

(q) $5x - 4y = 1$

(c) $\sqrt[3]{x^2} + \sqrt[3]{y^2} = 9$

(r) $x^2 + y^2 = 8$

(d) $x = 2t - t^2, y = t + t^2$

(s) $x^2 + y^2 = 6x - 1$

Choose the correct option :

(A) (a)-p; (b)-q; (c)-r; (d)-s

(B) (a)-s; (b)-r; (c)-p; (d)-q

(C) (a)-q; (b)-s; (c)-r; (d)-p

(D) (a)-r; (b)-p; (c)-s; (d)-q

Ans. B

Sol.

T03/5/282. If ΔABC is right angled triangle, right angled at vertex A(1, 0) and co-ordinates of other vertices are non-negative. The angle bisector through A is $x = 1$ and angle bisector through B is $(2 - \sqrt{3})x + y = 1$. Point B is such that slope of AB is less than slope of AC. Let I be incentre of ΔABC .

यदि ΔABC एक समकोण त्रिभुज है जहाँ समकोण वाला शीर्ष A(1, 0) है तथा दूसरे शीर्षों के निर्देशांक अऋणात्मक हैं। A से कोणार्धक का समीकरण $x = 1$ तथा B से कोणार्धक का समीकरण $(2 - \sqrt{3})x + y = 1$ है। बिन्दु B इस प्रकार है कि AB का ढाल AC के ढाल से कम है। माना I ΔABC का अन्तःकेन्द्र है।

Match the following column :

निम्न कॉलम का सुमेलित कीजिए :

Column-I

Column-II

(a) Coordinates of incentre I are

(p) $(1, \sqrt{3})$

(b) Co-ordinates of vertex B are

(q) $(1, \sqrt{3} - 1)$

(c) In ΔABC , if $\sin C = \alpha$ & $\sin B = \beta$, then ordered pair $(2\alpha, 2\beta)$ is given by

(r) (0, 1)

(d) Co-ordinates of vertex C are

(s) $(1 + \sqrt{3}, \sqrt{3})$

स्तम्भ -I

- (a) I के निर्देशांक होंगे
 (b) B के निर्देशांक होंगे
 (c) ΔABC में यदि $\sin C = \alpha$ तथा $\sin B = \beta$ है, तो क्रमित युग्म $(2\alpha, 2\beta)$ होगा
 (d) C के निर्देशांक होंगे

(t) $(1 - \sqrt{3}, \sqrt{3} - 2)$

स्तम्भ -II

- (p) $(1, \sqrt{3})$
 (q) $(1, \sqrt{3} - 1)$
 (r) $(0, 1)$
 (s) $(1 + \sqrt{3}, \sqrt{3})$
 (t) $(1 - \sqrt{3}, \sqrt{3} - 2)$

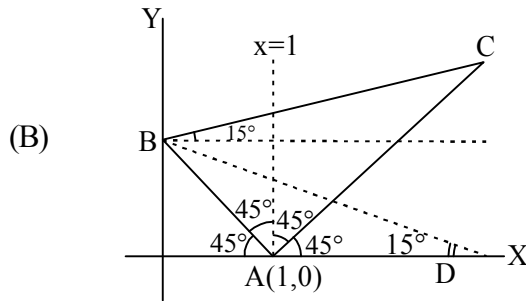
Choose the correct option :

सही विकल्प का चयन कीजिए :

- (A) (a) $\rightarrow$ (r); (b) $\rightarrow$ (q); (c) $\rightarrow$ (p); (d) $\rightarrow$ (s) (B) (a) $\rightarrow$ (q); (b) $\rightarrow$ (r); (c) $\rightarrow$ (p); (d) $\rightarrow$ (s)
 (C) (a) $\rightarrow$ (r); (b) $\rightarrow$ (p); (c) $\rightarrow$ (q); (d) $\rightarrow$ (s) (D) (a) $\rightarrow$ (r); (b) $\rightarrow$ (p); (c) $\rightarrow$ (s); (d) $\rightarrow$ (q)

Ans. B

Sol. (A) Incentre : intersection point of $x = 1$ and $(2 - \sqrt{3})x + y = 1$



$\therefore$ Slope of AB < Slope of AC

$$\therefore m_{AB} = \tan 135^\circ = 1$$

$$m_{AC} = \tan 45^\circ = 1$$

$$\Rightarrow \text{Line AB : } y = -1(x - 1) \Rightarrow x + y = 1 \text{ and Line AC : } y = x - 1$$

$\Rightarrow$ Point B is point of intersection of line AB and angle bisector through B

$$\text{i.e. } (2 - \sqrt{3})x + y = 1$$

$\Rightarrow$ Point B is $(0, 1)$

(C) $m_{BI} = -(2 - \sqrt{3}) = -\tan 15^\circ$

$$\Rightarrow \angle ADB = 15^\circ \text{ (see figure)}$$

$$\Rightarrow \angle ABD = 180^\circ - (135^\circ + 15^\circ) = 30^\circ$$

$$\Rightarrow \angle B = 60^\circ \text{ and } \Rightarrow \angle C = 30^\circ$$

$$\Rightarrow (2\alpha, 2\beta) \equiv (1, \sqrt{3})$$

$$(D) m_{BC} = \tan(15^\circ) = 2 - \sqrt{3}$$

$$\Rightarrow \text{line BC is } (y - 1) = (2 - \sqrt{3})(x - 0)$$

Point C is point of intersection of line BC and line AC ($y = x - 1$)